

PRODUCT CATALOGUE

PCI CABLES INDUSTRIES

PVT. LTD.

ISO 9001:2015 | ISO 14001:2015 | ISO 45001:2018 Certified



Introduction

PCI Cables is a trusted name in the manufacturing of high-quality PVC/XLPE insulated power and control cables, service cables, aerial bunched cables up to 1.1 kV, MVCC upto 11 kV, and all types of overhead conductors such as AAC, ACSR, AACSR, AAAC, AL-59, ACAR, ECO & HTLS conductor up to 800 KV. Committed to international standards since September 2013, PCI caters to the diverse needs of electricity boards across India and overseas markets, adhering to BIS, BS, IEC, ASTM, and other global benchmarks.

Wires and cables are the lifelines of modern infrastructure, enabling seamless electricity flow and powering every aspect of daily life—from planes to trains, telecommunications to entertainment, satellites to Wi-Fi. PCI Cables embodies this vital role, offering products that bring efficiency, safety, and cost savings with every application.

Renowned for quality and innovation, PCI's products are recognized with numerous certifications and awards, solidifying its reputation as one of India’s premier wire and cable manufacturers.

Expanding its horizons, PCI has ventured into the project division, undertaking EPC (Engineering, Procurement, and Construction) and turnkey projects in power transmission and distribution. While this is a new domain for PCI, the company has demonstrated a commitment to excellence, executing each project with precision, care, and unwavering dedication.

Our Mission

The company has established a state-of-the-art manufacturing setup with modern infrastructure, utilizing the latest machinery and equipment for production and testing. Our advanced laboratory is equipped with cutting-edge testing facilities to ensure the highest quality of manufactured products. We are committed to continuous and systematic efforts to achieve and maintain exceptional standards in quality and innovation.

Our Vision

We have effectively implemented a long-term growth strategy to achieve industry leadership in the cable and conductor sector. This approach focuses on continuous innovation, quality enhancement, and strategic market expansion, enabling the company to meet evolving customer needs and maintain a competitive edge.

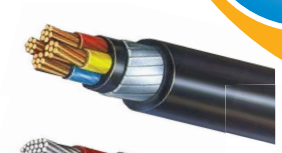
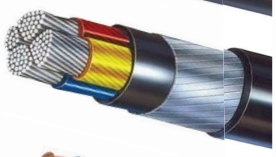
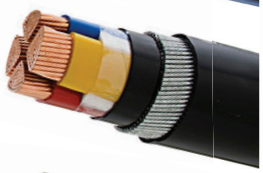
Our Objective

PCI Cables is committed to aligning with India's rapid economic growth by offering comprehensive end-to-end solutions and services. With a customer-centric approach, a global vision, and IT-driven processes that simplify business interactions, PCI Cables has earned the trust and respect of its esteemed clients.

Product Line

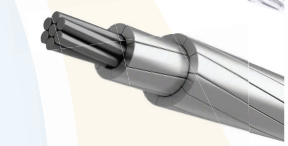
CABLES

- AERIAL BUNCHED CABLE
- LT XLPE CABLE
- PVC POWER CABLE
- CONTROL CABLE
- SERVICE CABLE
- COVERED CONDUCTORS FOR OVERHEAD LINES







CONDUCTORS

- ALUMINIUM ALLOY CONDUCTOR (AAC)
- ALL ALUMINUM ALLOY CONDUCTORS (AAAC)
- ALUMINUM CONDUCTOR STEEL REINFORCED (ACSR)
- ALUMINUM ALLOY CONDUCTOR STEEL REINFORCED (AACSR)
- ALUMINUM ALLOY CONDUCTOR 59 (AL-59)
- ALUMINUM CONDUCTOR ALLOY REINFORCED (ACAR)
- ECO CONDUCTOR

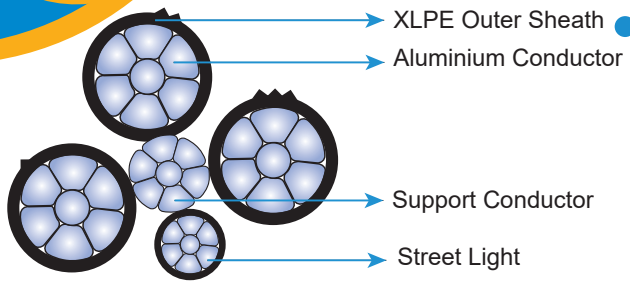
GI EARTH WIRES

- GI EARTH WIRE
- BEC (STEEL) UP TO 19 STRAND



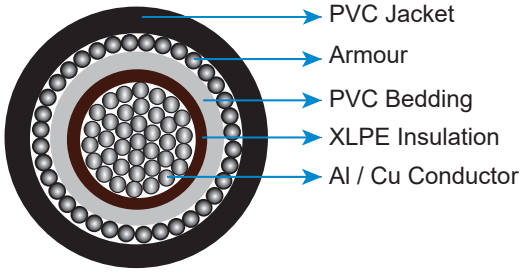
CABLES

Aerial Brunched Cable



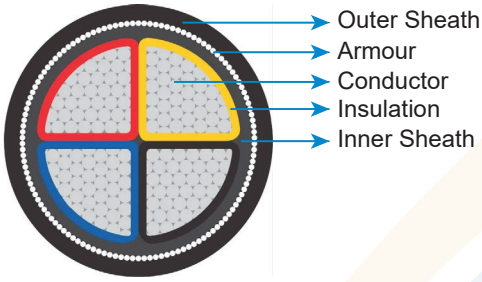
XLPE/PE insulated phase conductors.
One insulated Neutral conductor (if required) and a street lighting conductor (if required) shall be twisted around the bare or insulated messenger conductor.

LT XLPE Cable



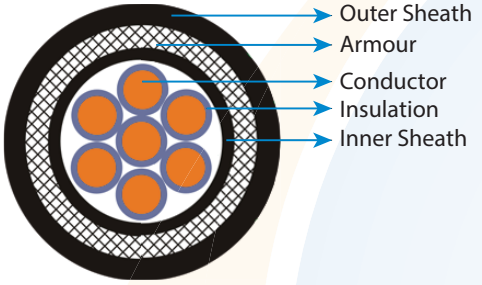
XLPE is an acronym for cross linked polyethylene. XLPE is a thermosetting polymer meaning that the material (polyethylene (PE)) is cured under heat and in that process forms bonds in all directions forming a three dimensional matrix.

PVC Power Cable



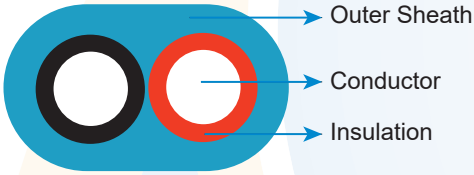
A PVC cable – meaning poly vinyl chloride – is an electrical cable that is coated with PVC, and offers superior insulation and protection to the main wire. PVC insulated cables provide property owners with several advantages: They have no chemical properties and hence, can widely be used in various chemical industries.

Control Cable



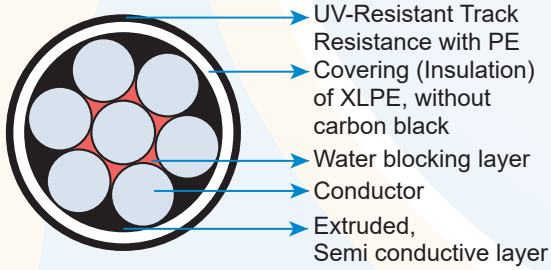
Control cables are multi-conductor cables used in automation and instrumentation applications. Control cables can measure and regulate transmissions of automated processes. Control cables are often UL rated. Control cables typically are shielded with a foil shield, braid shield or combination of the two.

Service Cable



Service Cable means that portion of the Company's cables, wires, ducts and conduits C connected to the main distribution cable and located on the premises of the applicant or customer for whom service is or will be supplied.

Covered Conductors for Overhead Lines

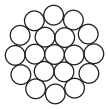


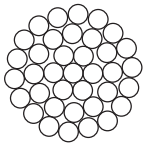
Covered conductors consist of a conductor surrounded by a covering made of insulating material as protection against accidental contacts with other covered conductors and with grounded parts such as tree branches, etc. In comparison with insulated conductors, this covering has reduced properties, but is able to withstand the phase-to-earth voltage temporarily. covered conductors for overhead lines having a nominal voltage U above 1 kV AC up to and including 36 kV AC.

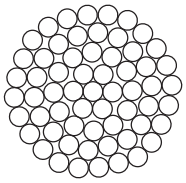
CONDUCTORS

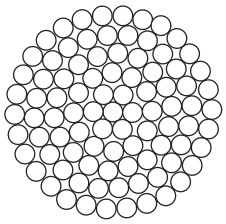
ALL ALUMINIUM CONDUCTOR (AAC)


7 Al Strand


19 Al Strand


37 Al Strand

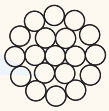

61 Al Strand

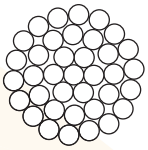

91 Al Strand

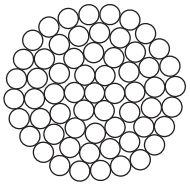
Conductor shall be all Aluminium stranded circular (as per IS-398 (Pt-1) or IEC 61089) or any other international Specificaion such as BS, ASTM, DIN, Canadian, GOST & Swedish specification etc.

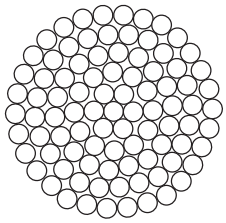
ALL ALUMINIUM ALLOY CONDUCTOR (AAAC)


7(1+6) Al Alloy


19 (1-6-12) Al Alloy

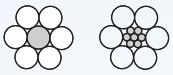

37 (1+6+12+18) Al Alloy


61 (1+6+12+18+24) Al Alloy

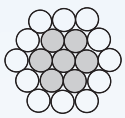

91 (1+6+12+18+24+30) Al Alloy

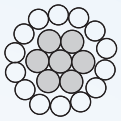
Conductor shall be Aluminium Magensium Silicon alloy strandard circular (as per IS -398 (Pt-4) or IEC 61089) or any other international specification, such as BS, ASTM, DIN, Canadian, GOST & Swedish specification etc.

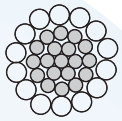
ALUMINIUM CONDUCTOR STEEL REINFORCED (ACSR)

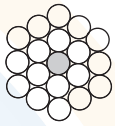

6+1

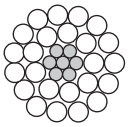
6+7(1+6)

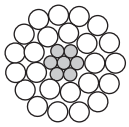

12+7(1+6)

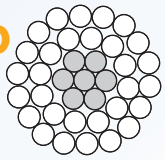

14+7(1+6)

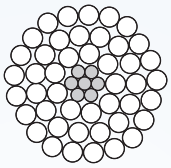

14+19(1+6+12)

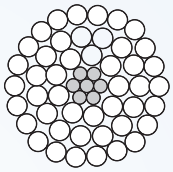

18(6+12)+1

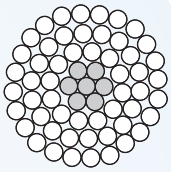

24(9+15)+7(1+6)

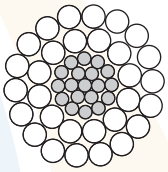

26(10+16)+7(1+6)

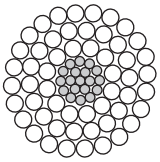

30(12+18) +7(1+6)


45(9+15+21) +7(1+6)


48(10+16+22) +7(1+6)


54(12+18+24) +7(1+6)

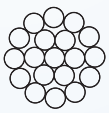

30(12+18) +19(1+6+12)

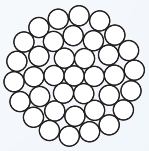

54(12+18+24) +19(1+6+12)

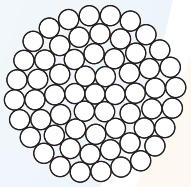
Conductor shall be Aluminium strandard circular with steel reinforced (as per IS-398 (Pt-2), IS-398 (Pt-5) or IEC 61089) or any other international specification, such as BS, ASTM, DIN, Canadian, GOST & Swedish specification etc.

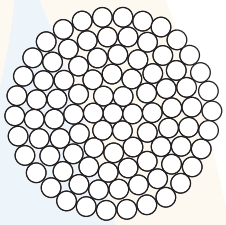
ALUMINIUM CONDUCTOR 59 (AL-59)


7(1+6)


19 (1-6-12)

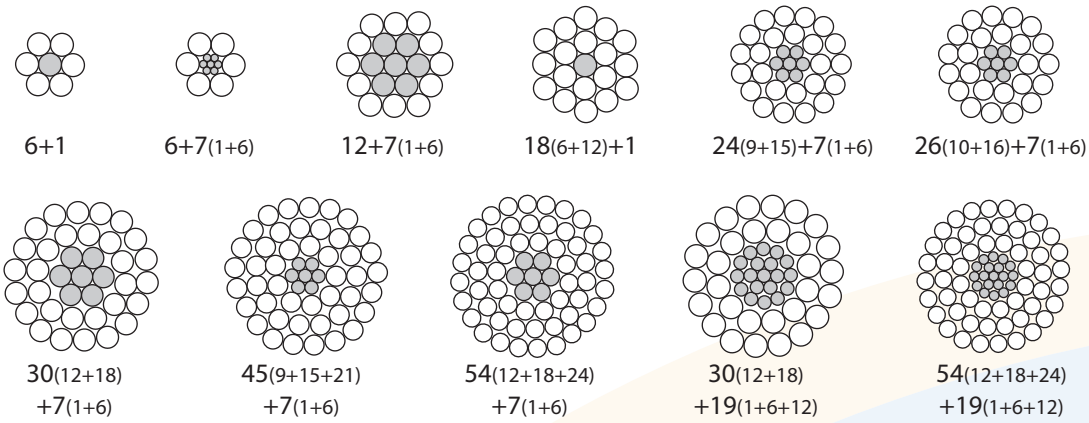

37 (1+6+12+18)


61 (1+6+12+18+24)


91 (1+6+12+18+24+30)

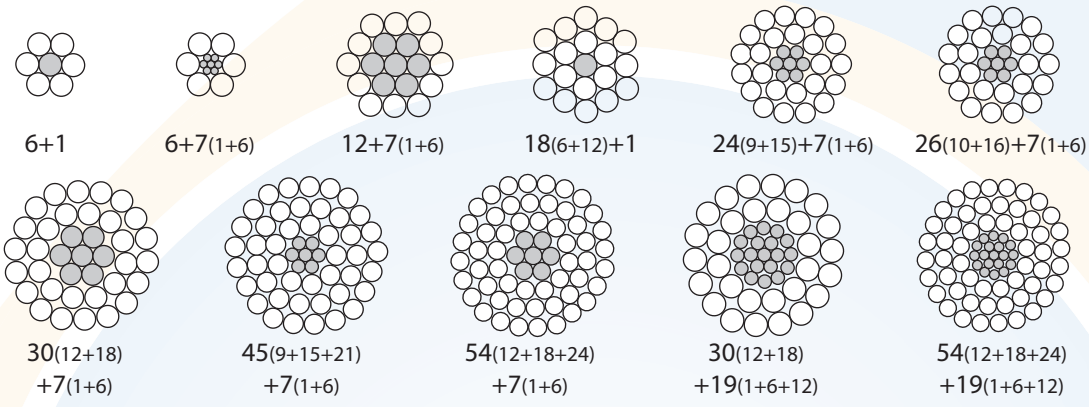
Aluminium Magnesium Silicon alloy with minimum 59% conductivity, as compared to ACSR conductor (as per IS-398 (Pt-6) or IEC 61089) or any other international standards, of equivalent diameter or weight is having higher current carrying capacity.

ALUMINIUM ALLOY CONDUCTOR STEEL REINFORCED (AACSR)



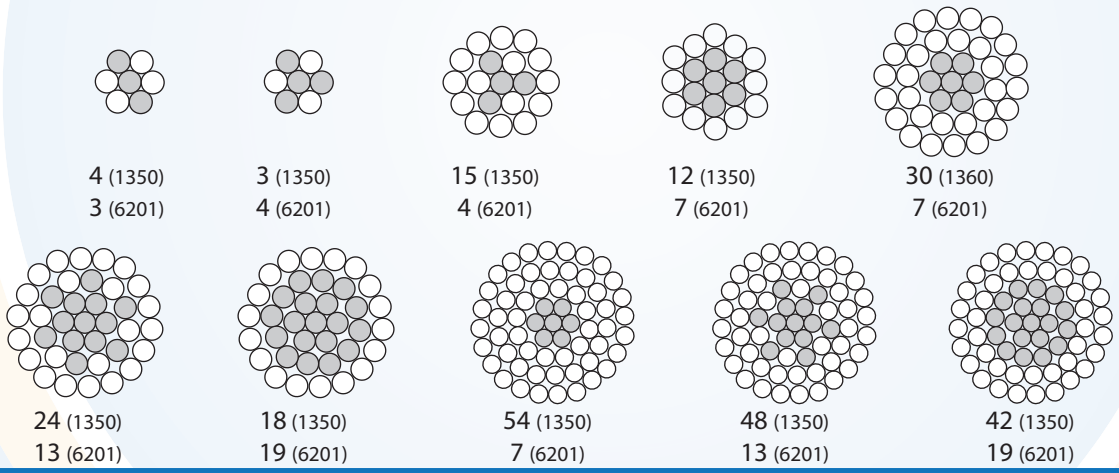
Conductor shall be Aluminium Magnesium alloy Silicon strandard circular with steel reinforced (as pe IEC 61089) or any other international specification, such as BS, ASTM, DIN, Canadian, GOST & Swedish specification etc.

THERMAL-RESISTANT ALLOY CONDUCTOR STEEL REINFORCED (TACSR)



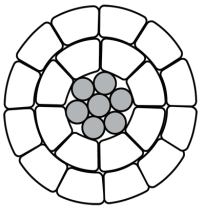
The core of the conductor is made of steel (galvanized or aluminum-clad), while the outer layer is composed of a thermal-resistant aluminum alloy, allowing it to operate at higher temperatures without significant loss of mechanical strength.

ALUMINIUM CONDUCTOR ALLOY REINFORCED (ACAR)



ACAR has got a better mechanical and electrical properties as compared to an equivalent ACSR, AAC or AAAC. A very good balance between the mechanical and electrical properties therefore makes ACAR the best choice where the ampacity, strength and light weight are the main consideration of the line design.

ECO CONDUCTOR

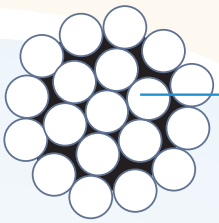


Al-59 Round Shape [7 (1+6)]+Trapezodial Al-59 [20 (8+12)]

Eco Series conductors consists of trapezoidal shaped special Aluminium Alloy (AL-59) containing Magnesium, Silicon and Copper stranded with round core. The trapezoidal shape provides up to 96% compactness.

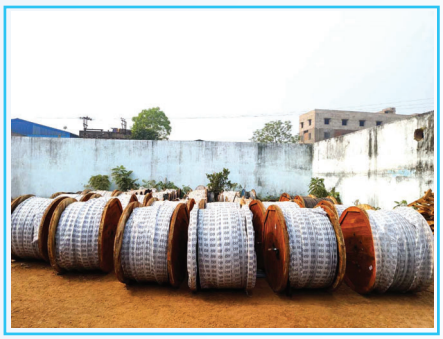
EARTH WIRE

GI EARTH WIRE



Galvanized High Corbon Steel Wire

Earth wires are made from stranded galvanized high carbon steel wires. They comprise of 3 to 19 strands of such wires. All steel strands are smooth, uniform and free from imperfections such as die-marks, cents after drawing and standing, scratches and abrasion.













ALL ALUMINIUM CONDUCTORS (AAC)

AS PER IS-398-1: 1996

Code Word	Nominal Aluminium Area	Stranding and Wire Diameter	Sectional Area	Approximate Overall Diameter	Approximate Mass	Calculated Resistance at 20C, Max	Approximate Calculated Breading Load
	Sq. / mm	No. / mm	Sq. / mm	mm	Kg / km	Ω / km	KN
GNAT	25	7/2.21	26.85	6.63	74	1.096	4.52
ANT	50	7/3.10	52.83	9.3	145	0.5525	8.25
WASP	100	7/4.39	106	13.17	290	0.2752	15.96
HORNET	150	19/3.25	157.6	16.3	434	0.1825	27.7
SPIDER	240	19/3.99	237.6	19.95	654	0.1235	35.74
BUTTERFLY	300	19/4.65	322.7	23.25	888	0.09107	48.74

AS PER BRITISH SPECIFICATION BS-215-I-1970

Code Word	Conductor Size sq. mm	Standing & Wire dia no./mm	Sectional Area sq. mm	Overall Diameter mm	Approx. Weight Kg/Km	Ultimate Strength Kg	Upper Resistance at 20°c (Max.) ohm/Km
MIDGE	22	7/2.06	23.33	6.18	64	407	1.227
GNAT	25	7/2.21	26.85	6.63	74	468	1.066
ANT	50	7/3.10	52.83	9.3	145	84	0.5419
FLY	60	7/3.40	63.55	10.2	174	1010	0.4505
WASP	100	7/4.39	106	13.17	290	1632	0.2702
HORNET	150	19/3.25	157.6	16.25	434	2620	0.1825
CHAFER	200	19/3.78	213.2	18.9	587	3300	0.1349
COCKROACH	250	19/4.22	265.7	21.1	731	4120	0.1033
BUTTERFLY	300	19/4.65	322.7	23.25	888	4976	0.08916
CENTIPEDE	400	37/3.78	415.2	26.46	1145	6430	0.06944
TARANTULA	795	37/5.23	794.8	36.6	2191	12014	0.03628
BULL	865	61/4.25	864.92	38.25	2400	14169	0.03339

AS PER AMERICAN SPECIFICATION ASTM B-231 - 2016

Code Word	Conductor Size cmils/aWg	Stranding & Wire dia no./mm	Sectional Area sq. mm	Overall Diameter mm	Approx. Weight Kg/Km	Ultimate Strength Kg	Approx. Resistance at 20°c (Max.) ohm/Km
PANSY	(1)	712.78	42.39	8.32	116.8	746	0.6752
POPPY	(1/10)	7/3.12	53.48	9.4	147.4	900	0.5351
ASTER	(2/0)	7/3.50	67.42	10.51	185.8	1140	0.4245
LILY	(3)	7/2.207	26.66	6.606	73.5	515	1.076
OXLIP	(4/0)	7/4.41	107.2	13.3	295.6	1840	0.2671
TULIP	336400	1413.18	170.5	16.91	470	21290	0.169
HYACINTH	500000	3712.95	753.1	20.7	698.5	11.111	0.1138
ORCHID	636000	37/3.33	322.2	23.11	888.4	51511	0.0896
LILAC	795000	61/2.90	403.2	26.11	1112	6510	0.0719
JESSAMINE	1750000	61/4.30	886.9	38.73	2445	13500	0.0326
COWSLIP	2000000	91/3.76	1013	41.4	2793	13600	0.0284

AS PER GERMAN SPECIFICATION DIN 48 201 TEIL 5/1981

Nominal Aluminium Area sq. mm	Stranding & Wire dia no./mm	Approx. Overall Diameter mm	Sectional Area sq. mm	Approx Weight Kg/ Km	Approx. Calculated Breaking Load Kn	Calculated Resistance at 20°c (Max.) ohm/Km
16	711.70	5.10	15.89	43	2„84	I 8017
25	717.10	6.30	24.25	66	*4* 17	1 1806
35	7/2.50	7.50	34.36	94	5 78	0 8332
50	7/3 00	9.00	49.48	135	7.94	0.5786
50	19/1 80	9.00	48 35	133	8.44	0 5948
70	19/2.10	10.50	65.81	181	11.32	0.4370
95	1912.50	12.50	93.27	256	15.67	0.3084
120	19/2.80	14.00	117.00	322	18.77	0.2459
150	37/2.25	15.80	147.11	406	25:29	0.1960
185	37/2.50	17.50	181.62	500	30.54	0.1588
240	61/2.25	20.30	242.54	670	39.40	0.11920
300	61/2.50	22.50	299.43	827	47.69	0.09651
400	61/2.89	26.00	400.14	1104	60.86	0.07222
500	61/3.23	29.10	499.83	1379	74.67	0.05782
625	91/2.96	32.60	626.20	1732	95.24	0.04625
800	91/3.35	36.90	802.09	2218	118.39	0.03611
1000	91/3.74	41.10	999.71	2767	145.75	0.02897

ALL ALUMINIUM CONDUCTORS (AAC)

AS PER CANADIAN SPECIFICATION CAN/CSA - C61089 W11

Code Word	Stranding & Wire dia no./mm	Approx Overall Diameter mm	Sectional Area sq. mm	Approx. Weight Kg/Km	Approx. Calculated Breaking load Kn	Calculated Resistance at 20°c (Max.) ohm/Km
ROSE	7/1.96	5.89	21.16	58	4.07	1.351
LILY	7/2.20	6.61	26.65	73	5.05	1.072
IRIS	7/2.47	7.42	33.61	92	6.23	0.8497
PANSY	7/2.78	8.33	42.39	116	7.6	0.6739
POPPY	7/3.12	9.36	53.48	146	9.22	5.5341
ASTER	7/3.50	10.51	67.47	184	11.62	0.4236
PHLOX	7/3.93	11.8	85.03	232	14.07	0.336
OXLIP	7/4.42	13 25	107.23	293	17.75	0.2664
DALSY	7/4.96	14.9	135.16	369	22.36	0.2113
VALERIAN	19/2.91	14.57	126.71	348	22.26	0.2274
LAUREL	19/3.01	15.05	135.16	371	23.78	0.2129
PEONY	19/3.19	15.97	152	417	26.15	0.189
TULIP	19/3.38	16.11	170.45	467	29.3.8	0.1638
DAFFODIL	19/3.45	17.24	177.35	488	30.56	0.1624
CANNA	19/3.67	36.16	201.41	554	34.04	0.1427
GOLDTUFT	19/3.91	19.55	228	626	37.72	0.1263
COSMOS	19/4.02	20.12	241.68	664	40.07	0.1166
ZINNIA	19/4.12	20.6	251.35	695	41.91	0.1137
DAHLIA	19/4.35	21.73	282	774	46.69	5.1018
MEADOWSWEET	37/3.23	22.63	304	838	52.43	0.0948
ORCHID	37/3.33	23.11	372.26	688	55.57	0.0896
HEUCHERA	17/3.32	23.56	329.35	906	56.79	0.0876
VERBENA	37/3.49	24.45	354.77	978	61.16	8.0814
VIOLET	37/3.53	24:74	362.58	1000	62.34	0.0797
PETUNIA	37/3.62	48341	380	1048	64.2	0.0758
ARBIJRIS	37/3.72	26.06	402.84	1112	68.02	0.0715
ANEMONE	37/3.90	27.33	443.1	1222	73.32	0.0653
COCKSOMB	37/3.96	27.73	456.06	1257	75.48	0.0633
MAGNOLIA	37/4.06	28.55	483.42	1333	80	4.0597
HAMMED	37/4.18	29.23	506.71	1396	83.87	0.0568
BLUEBELL	37/4.24	29.72	523.68	1415	86.48	0.0551
MARIGOLD	61/3.43	30.89	563.93	1559	97.22	0.0512
HAWTHORN	61/3.55	31.95	604.26	1670	104.13	0.0479
NA.C15505	61/3.67	33.02	645.37	1781	108.79	0.045
COLUMBINE	61/3.78	34.01	684.84	1893	115.71	0.0423
CARNATION	61/389	35.03	725.1	2004	119.92	0.04
GLADIOLUS	61/4.00	35.99	765.35	2116	126.59	0.0377
COREOPSIE	61/4.10	36.91	205.68	2226	133.26	0.0358

ALUMINIUM CONDUCTORS STEEL REINFORCED (ACSR)

AS PER IS-398-II: 1996 & IS-398-V: 1992

Code Word	Nominal Aluminium Area sq. mm	Stranding & Wire Dia no./mm Alum.	Steel	Sectional Area (Aluminium) sq. mm	Total Sectional sq. mm	Approx. Overall mm	Approx. Weight Kg/Km Total	Alum.	Steel	Calculated at 20°c (Max.) ohm/Km	Approx. Calculated Breaking Load Kn
MOLE	10	6/1.50	1/1.50	10.6	12.37	4.5	43	29	14	2.78	3.97
SPL. CONDUCTOR	18	6/1.96	1/1.96	18.1	21.12	5.88	73	50	23	1.618	6.74
SQUIRREL	20	6/2.11	1/2.11	20.98	24.48	6.33	85	58	27	1.394	7.61
WEASEL	30	6/2.59	1/2.59	31.61	36.88	7.77	128	87	41	0.9289	11.12
RABBIT	50	6/3.35	1/3.35	52.88	61.7	10.05	214	145	69	0.5524	18.25
RACCOON	80	6/4.09	1/4.09	78.83	91.97	12.27	319	217	102	0.3712	26.91
DOG	100	6/4.72	7/1.57	105	118 50	14.15	394	288	106	0.2792	32.41
WOLF	150	30/2.59	7/2.59	158.1	194.9	18.13	726	438	288	0.1871	67.34
PANTHER	200	30/3.00	7/3.00	212.1	261.5	21	974	587	387	0.1391	89.67
KUNDAH	400	42/3.50	7/1.96	404.1	425.2	26.88	1281	1120	161	0.07312	88.79
ZEBRA	420	54/3.18	713.18	428.9	484.5	28.62	1621	1187	434	0.06868	130.32
MOOSE	520	54/3.53	7/3.53	528.5	597	31.77	2004	1465	539	0.05595	159.6
MARKULLA	560	42/4.13	7/2.30	562.7	591.7	31.68	1781	1559	228	0.05231	120.16
BERSIMIS	690	42/4.57	7/2.54	688.9	724.4	35.04	2187	1907	280	0.04242	146.87

ALUMINIUM CONDUCTORS STEEL REINFORCED (ACSR)

AS PER BRITISH SPECIFICATION BS-215-II-1970

acsr code World	conductor) size sq. mm	stranding & wire dia no./mm		calculated sectional area sq mm		overall diameter mm	approx Weight kg/km			ultimate strength Kg	calculated resistance at 20°c (Max.) ohm/Km
		alum	steel	alum	acsr	acsr	acsr	alum	steel		
GOPHER	25	6/2.36	1/2.36	26.24	30.62	7.08	106	72	34	980	1.093
WEASELL	30	6/2.59	1/2.59	31.61	36.58	7.77	128	87	41	1170	0.9077
FOX	35	6/2.79	1/2.79	36.48	42.8	8.37	146	101	47	1340	0.7822
FERRET	40	6/3.00	1/3.00	42 41	49.43	9	172	117	55	1550	0.6766
RABBIT	50	6/3.35	1/3.35	52.88	61.7	10.05	214	145	69	1870	0.5426
MINK	60	6/3.66	1/3.66	63.17	72.64	10.98	255	173	82	2220	0.4546
HORSE	70	12/2.79	7/2.79	73.37	116.2	13.95	538	203	335	6240	0.3936
DOG	100	6/4.72	7/1.57	105	118.5	14.15	394	288	106	3330	0.2733
COYOTE	130	2672.54	7/1.91	131.7	151.8	15.89	521	364	157	4730	0.2191
WOLF	150	30/2.59	7/2.59	158.1	194.9	18.13	726	437	289	7060	0.1828
LYNX	175	30/2.79	7/2.79	183.4	226 20	19.53	642	507	335	8140	0.1576
PANTHER	200	30/3.00	7/3.00	212.1	261.5	21	974	586	388	9430	0.1363
JAGUAR	200	18/3.86	7/3.86	210.6	222.3	19.3	671	580	91	4750	0.1367
GOAT	320	30/3.71	7/3.71	324.3	400	25.97	1489	896	593	13850	0.08912
BISON	380	54/3.00	7/3.00	381.7	431.2	27	1443	1055	886	12330	0.07574
ZEBRA	400	54/3.18	7/3.18	428.9	484.5	28.62	1622	1186	436	13450	0.0674
DEER	425	30/4.27	7/4.27	429.6	529.8	29.89	1973	1187	786	16210	0.06727
MOOSE	525	54/3.53	7/3.53	528.5	597	31.71	1998	1461	537	16420	0.0547

AS PER GERMAN SPECIFICATION DIN 48.204/1974

Nominal Size sq. mm	Stranding & Wire Dia no./mm		Approx. Overall Diameter mm	Sectional Area sq. mm		Total Sectional Area sq. mm	Approx. Weight Kg/Km			Approx. Calculated Breaking Load Kn	Calculated Resistance at 20°c (Max) ohm/Km
	Alum	Steel		Alu.	Steel		ACSR	Alum	Steel		
16/2.5	6/1.80	1/1.80	5.4	15.3	2.55	17.85	42	20	62	5.83	1.875
25/4.0	6/2.25	1/2.25	6.8	23.8	4	27.8	65	32	97	9.02	1.206
35/6.0	6/2.70	1/2.70	8.1	34 30	5.7	40	94	46	140	!2_41	0.8365
44/32	14/2.0	7/2.40	8.2	44	31.7	75.7	122	250	372	44 i4	0.657
50/8	6/3.20	1/3.20	9.6	48.3	5	56 30	137	64	196	16.77	0.5941
50/30	12/2.3	7/2.33	11.65	51.2	29.8	8 l .00	141	237	378	42_96	0.5642
70/12	26/1.5	7/1.44	11.7	69.9	11.4	81.3	193	91	284	16.29	0.113
95/15	26/2.5	7/1.67	13.6	94.4	15.3	109 70	260	123	383	35.07	0.3058
95/55	12/3.0	7/3.20	16	96.5	56.3	152.8	266	446	712	77.84	0.2992
105/75	14/3.0	19/2.5	17.5	105.7	75.5	181.5	292	599	891	106 38	0.2735
120/20	26/2.4	7/1.90	15.5	121.6	19.8	141.4	336	158	494	4.78	0.2741
120/70	12/3.0	7/3.60	18	122	71.3	193.3	337	564	901	98.1	0.2367
125/30	30/2.3	7/2.33	16.1	127.9	29.8	157.7	353	138	511	56.5	0.2259
150/25	26/20	7/2.10	17.1	148.9	24.2	171.1	411	194	605	54.2	0.1939
170/40	30/2.0	7/2.70	18.9	71.8	40.1	211.9	475	319	794	75.29	0.1682
185/30	26/3.0	7/2.33	19	183 80	29.5	213.6	507	239	146	64.94	0.15/1
210/35	26/3.0	7/2.49	20.3	209.1	14.1	243.2	577	273	850	73.47	0.138
210/50	30/3.0	7/3.00	21	212.1	49.5	261.6	587	94	981	92.11	1.1362
230/30	24/3.0	7/2.33	21	230.9	29.8	260.7	638	239	877	7.16	0.1249
240/40	26/3.5	7/2.68	21.9	243	39.5	282.5	671	316	987	84.7	0.1188
265/35	24/3.4	7/2.49	22.4	263.7	31.1	797.8	728	274	1007	81.47	0.1094
300/50	26/3.6	7/3.00	24.5	304.3	49.5	353.8	840	396	1236	104.96	0.09486
305/40	54/2.8	7/2.68	24.1	304.6	39.5	344.1	843	317	1160	97.51	0.0949
340/30	48/3.0	7/2.33	25	339.3	29.6	369.1	938	242	1180	91.13	0.08508
380/50	54/3.0	7/3.00	27	382	49.5	431.5	1056	397	1453	120.7	0.07568
385/35	48/3.0	7/2.49	26.7	386	34.1	420.1	1067	277	1344	102.8	0.07479
435/55	54/3.0	7/3.20	28.8	434.3	56.3	490.6	1203	450	1653	133.8	0.06686
450/40	48/3.5	7/2.68	28.7	448.7	39.5	489.2	1241	320	1561	118.45	0.06434
490/65	54/3.0	7/3.40	30.6	490.3	63.6	553.9	1356	510	1866	150.19	0.05896
495/35	45/3.4	7/2.49	29.9	494.1	34.1	528.2	1369	2.77	1646	119.5	0.05846
510/45	48/3.8	7/2.87	30.7	510.2	45.3	555.5	1413	365	1778	134	0.05661
550/70	4/3.0	7/3.60	32.4	550	71.3	621.3	1520	572	2092	167.35	0.05256
560/50	48/3.6	7/3.00	32.2	561.7	49.5	611.2	1553	401	1954	146.12	0.0514
570/40	45/4.2	7/2.68	32.2	571.2	39.5	610.7	1571	317	1888	133.61	0.05057
650/45	45/4.0	7/2.87	34	653.5	45.3	698.8	1806	365	2171	152.54	0.0442
680/85	54/4.0	19/2.0	36	678.6	86	764.6	1868	702	2570	202.33	0.04259
1045/45	72/4.0	7/2.87	43	1045.6	45.3	1090.9	2893	358	3251	213.46	0.02765

ALUMINIUM CONDUCTORS STEEL REINFORCED (ACSR)

AS PER CANADIAN SPECIFICATION CAN/CSA - C61089 - 11

Code Name	Stranding & Wire Dia no./mm		Approx Overall Diameter mm	Sectional Area sq. mm		Total Sectional Area sq. mm	Approx Weight Kg/Km			Approx. Calculated Breaking Load Kn	Calculated Resistance at 20°c (Max.) ohm/Km
	Alum.	Steel		Alu.	Steel		Alu.	Steel	ACSR (TOTAL)		
WREN	6/1.33	1/1.33	3.99	8.39	1.42	9.81	23	11	34	3.33	3.4280
WRABLER	6/1.50	1/1.50	4.50	10.59	1.77	12.36	29	14	43	4.16	2.7200
TURKEY	6/1.68	1/1.68	5.04	13.29	2.19	15.48	37	17	54	5.19	2.1570
THRUSH	6/1.89	1/1.89	5.67	16.77	2.77	19.54	46	22	68	6.47	1.7711
SWAN	6/2.12	1/2.12	6.36	21.16	3.55	24.71	58	27	85	8.14	1.3570
SWALLOW	6/2.38	1/2.38	7.14	26.65	4.45	31.09	73	35	108	10.05	1.0760
SPARROW	6/2.67	1/2.67	8.01	33.61	5.61	39.22	92	44	136	12.44	0.8532
ROBIN	6/3.00	1/3.00	9.00	42.39	7.10	49.49	116	55	171	15.54	0.6766
RAVEN	6/3.37	1/3.37	10.11	53.48	8.90	62.38	146	69	215	18.90	0.5362
QUALL	6/3.78	1/3.78	11.34	67.42	11.23	78.65	185	88	273	23.50	0.4255
PIGEON	6/4.25	1/4.25	12.75	85.03	14.19	99.22	233	110	343	29.60	0.3375
PENGUIN	6/4.77	1/4.77	14.31	107.23	17.87	125.10	294	139	433	37.47	0.2677
OWL	6/5.36	7/1.79	16.09	135.16	17.55	152.71	371	137	508	42.10	0.2123
WAXWING	18/3.9	1/3.09	15.45	135.16	7.48	142.54	372	58	430	31.20	0.2130
PARTRIDGE	26/2.57	7/2.00	16.28	135.16	22.00	157.16	374	171	545	50.03	0.2136
PHOEBE	18/3.28	1/3.28	16.40	152.00	8.45	160.45	418	65	483	35.10	0.1895
OSTRICH	26/2.73	7/2.12	17.28	152.00	24.71	176.71	420	193	613	56.71	0.1900
PIPER	30/2.54	7/2.54	17.78	152.00	35.48	187.48	420	277	697	68.67	0.1900
MERLIN	18/3.47	1/3.47	17.35	170.45	9.48	179.93	469	74	54	39.40	0.1686
LINNET	26/2.89	7/2.25	18.31	170.45	27.81	198.26	470	217	687	62.30	0.1694
IBIS	26/3.14	7/2.44	19.85	201.42	32.77	234.19	557	256	813	72.00	0.1434
LARK	30/2.92	7/2.92	20.44	201.42	46.97	248.39	557	366	923	88.87	0.1434
PELICAN	18/4.14	1/4.14	20.70	241.68	13.42	255.10	665	104	769	54.78	0.1191
HAWK	26/3.44	7/2.67	21.77	241.68	39.42	281.10	667	308	975	86.52	0.1195
HEN	30/3.20	7/3.20	22.40	241.68	56.39	298.07	688	440	1128	103.88	0.1195
HERON	30/3.28	7/3.28	22.96	253.35	59.10	312.45	701	461	1162	108.79	0.1142
SAPSUCHER	22/4.04	7/2.24	22.85	282.00	27.68	309.68	777	216	993	79.11	0.1024
DOVE	26/3.72	7/2.89	23.55	282.00	45.94	327.94	778	359	1137	99.91	0.1024
EAGLE	30/3.46	7/3.46	24.22	282.00	65.81	347.81	780	513	1293	121.25	0.1027
DUCK	54/2.69	7/2.69	24.21	306.58	39.81	346.39	848	311	1159	100.11	0.0945
GROSBBEAK	26/3.97	7/3.09	25.15	322.26	52.45	374.71	890	409	1299	111.24	0.0896
EGRET	30/3.70	19/2.2	25.90	322.26	41.74	364.00	891	326	1217	105.26	0.0899
GOOSE	54/2.76	7/2.76	24.84	322.26	41.74	364.00	891	326	1217	105.26	0.0899
GULL	54/2.82	7/2.82	25.38	337.74	43.81	381.55	935	342	277	109.23	0.0856
STARLING	26/4.1	7/3.28	26.68	362.58	59.03	421.61	1002	460	1462	125.02	0.0796
REDWING	30/3.92	19/2.35	27.43	362.58	82.58	445.16	1002	646	1648	153.96	0.0796
CROW	54/2.92	7/2.92	26.28	362.58	46.97	409.55	1003	366	1369	117.22	0.0798
DRAKE	26/4.4	7/3.45	28.11	402.84	65.61	468.45	1112	512	1624	139.05	0.0715
MALLARD	30/4.14	19/2.48	28.96	402.84	91.87	494.71	1113	719	1832	171.08	0.0719
CONDOR	54/3.08	7/3.08	27.72	402.84	52.19	455.03	1113	408	1521	127.03	0.0719
CRANE	54/3.23	7/3.23	29.07	443.10	57.48	500.58	1226	448	1674	139.69	0.0653
CANARY	54/3.28	7/3.28	29.52	456.06	59.10	515.16	1262	462	1724	143.71	0.0633
CARDINAL	54/3.38	7/3.38	30.42	483.42	62.65	546.07	1338	488	1826	152.39	0.0597
CURLEW	54/3.51	7/3.51	31.59	523.68	67.87	591.55	1448	530	1978	165.29	0.0551
FINCH	54/3.65	19/2.19	32.85	563.93	71.55	636.48	1561	560	2121	176.88	0.0512
GRACKEE	54/3.77	19/2.27	33.97	604.26	76.58	680.84	1671	600	2271	191.78	0.0479
PHEASANT	54/3.90	19/2.34	35.10	644.51	81.68	726.19	1783	538	2421	199.33	0.0449
MARTIN	54/4.02	19/2.41	36.17	684.84	86.71	771.55	1894	679	2573	211.79	0.0432
PLOVER	54/4.14	19/2.48	37.24	725.10	91.87	816.97	2006	719	2725	224.25	0.0400
PARROT	54/4.25	19/2.55	38.25	765.35	96.84	862.19	2118	759	2877	237.15	0.0377
FALCON	54/4.36	19/2.62	39.26	805.68	102.13	907.81	2229	799	3028	249.61	0.0358

AS PER AMERICAN SPECIFICATIONS ASTM B - 232 - 1992

ACSR Code Word	Conductor Size (MCM/Awg)	Stranding & Wire Dia no./mm		Calculated Sectional Area sq.mm		Overall Diameter mm ACSR	Approx. Weight. Kg/Km			Ultimate Strength Kg	Calculated Resistance at 20°c (Max.) ohm/Km
		Alum.	Steel	Alum.	ACSR		ACSR	Alum.	Steel		
SPARROW	(2)	6/2.67	1/2.67	33.61	39.22	8.01	136	92	44	1295	0.853
SWALLOW	(3)	6/2.38	1 /2.38	26.65	31.1	7.14	108	73	35	1040	1.076
RAVEN	(1/0)	6/3.37	1/3.37	53.48	62.38	10.11	216	147	69	1990	0.5364
PENGUIN	(4/0)	6/4.77	1/4.77	107.23	125.1	14.31	433	294	139	3800	0.2676
MERLIN	336400	18/3.47	1/3.47	170.45	179.9	17.36	544	470	74	3930	0.1692
IBIS	397500	26/3.14	7/2.44	201.4	234	19.88	814	558	256	7380	0.1438
PELICAN	477000	18/4.14	1/4.14	241.68	255.1	20.7	771	666	105	5350	0.1193
HAWK	477000	26/3.43	7/2.67	241.7	280.8	21.78	976	669.2	307	8880	0.1194
GROSBEAK	636000	26/3.97	7/3.08	322.26	374.78	25.15	1303	893	409.8	11500	0.08989
DRAKE	795000	26/ 4.44	7/3.45	402.8	466.84	28.11	1628	1110	512	14300	0.07192
COOT	795000	36/3.77	1/3.77	402.8	414	26.42	1197	1110	87	7610	0.07134
MALLARD	795000	30/4.14	19/2.48	402.84	494.71	28.96	1839	1119	719	17450	0.07208
RAIL	954000	45/3.70	7/2.47	483.42	516.84	29.61	1600	1339	260	12020	0.05994
CARDINAL	954000	54/3.38	7/3.38	483.42	546.07	30.42	1829	1339	490	15500	0.05994
CURLEW	1033500	54/3.52	7/3.52	523.68	591.55	31.68	1981	1451	530	16700	0.05531
FALCON	1590000	54/4.36	19/2.62	805.8	908.12	39.26	3042	2243	799	25000	0.03612

ALUMINIUM CONDUCTORS STEEL REINFORCED (ACSR)

AS PER EN 50182:2001 - CHARACTERISTICS OF ACSR USED IN SWEDEN - TYPE AL1/ST1A

Code	Old Code	Area			No. of Wires		Wire Diameter		Diameter		Mass per unit length	Rated Strength	DC Resistance
		Al	Steel	Total			Al	Steel	Core	Cond.			
		mm^2	mm^2	mm^2	Al	Steel	mm	mm	mm	mm	kg/km	kN	Ohm/ kM
54-AL1/9-ST1A	RAVEN	53.5	8.92	62.4	6	1	3.37	3.37	3.37	10.1	216.1	18.64	0.5355
85-AL1/14-ST1A	PIGEON	85.1	14.2	99.3	6	1	4.25	4.25	4.25	12.8	343.6	29.22	0.3367
135-AL1/22-ST1A	PARTRIDGE	134.9	22.0	156.9	26	7	2.57	2.00	6.00	16.3	544.5	48.66	0.2141
201-AL1/33-ST1A	IBIS	201.3	32.7	234.1	26	7	3.14	2.44	7.32	19.9	812.1	70.53	0.1434
283-AL1/46-ST1A	DOVE	282.6	45.9	328.5	26	7	3.72	2.89	8.67	23.6	1139.6	97.56	0.1022
402-AL1/52-ST1A	CONDOR	402.3	52.2	454.5	54	7	3.08	3.08	9.24	27.7	1520.5	123.75	0.0719
525-AL1/52-ST1A	CURLEW	525.5	68.1	593.6	54	7	3.52	3.52	10.6	31.7	1986.0	159.01	0.0550
685-AL1/89-ST1A	SKATA	685.4	88.8	774.2	54	7	4.02	4.02	12.1	36.2	2590.2	207.39	0.0422
806-AL1/102-ST1A	FALCON	806.2	102.4	908.7	54	19	4.36	2.82	13.1	39.2	3031.7	245.77	0.0358
563-AL1/29-ST1A	MORKULLA	562.7	29.3	592.0	42	7	4.13	2.31	6.93	31.7	1783.7	123.47	0.0513
735-AL1/38-ST1A	RIPA	734.9	38.0	772.9	42	7	4.72	2.63	7.89	36.2	2327.5	160.93	0.0393
865-AL1/44-ST1A	ORRE	864.7	44.0	908.8	42	7	5.12	2.83	8.49	39.2	2733.1	188.55	0.0334
89-AL1/52-ST1A	DOTTEREL	89.4	52.2	141.6	12	7	3.08	3.08	9.24	15.4	654.8	72.12	0.3230
117-AL1/68-ST1A	ODEN	116.8	68.1	184.9	12	7	3.52	3.52	10.6	17.6	855.2	93.62	0.2473
152-AL1/89-ST1A	ATLE	152.3	88.8	241.2	12	7	4.02	4.02	12.1	20.1	1115.4	122.10	0.1895
251-AL1/65-ST1A	YMER	251.0	64.7	315.6	32	7	3.16	3.43	10.3	22.9	1199.8	112.56	0.1152

NOTE : Direction of lay of external layer is right-hand (Z)

AS PER EN 50182:2001 - CHARACTERISTICS OF ACSR USED IN FRANCE - TYPE AL1/ST1A

Code	Old Code	Area			No. of Wires		Wire Diameter		Diameter		Mass per unit length	Related Strength	DC Resistance	Final Modulus of	Coefficient of Liner expansion	Direction of lay of external layer
		Al	Steel	Total			Al	Steel	Core	Cond.						
		mm^2	mm^2	mm^2	Al	Steel	mm	mm	mm	mm	kg/km	kN	Ohm/ kM	N/mm2	1/K	
28-SL1/9-ST1A	CANNA 37.7	28.3	9.42	37.7	9	3	2.00	2.00	4.30	8.30	151.5	16.26	1.0187	88000	1.71E-05	S
38-SL1/22-ST1A	CANNA 59.7	37.7	22.0	59.7	12	7	2.00	2.00	6.00	10.00	276.1	32.7	0.766	103500	1.54E-05	S
48-SL1/28-ST1A	CANNA 75.5	47.7	27.8	75.5	12	7	2.25	2.25	6.75	11.30	349.4	41.15	0.6052	103500	1.54E-05	S
59-SL1/34-ST1A	CANNA 93.3	58.9	34.4	93.3	12	7	2.50	2.50	7.50	12.50	431.4	49.48	0.4902	103500	1.54E-05	Z
94-SL1/22-ST1A	CANNA 116.2	94.2	22.0	116.2	30	7	2.00	2.00	6.00	14.00	432.5	43.17	0.3067	75500	1.80E-05	S
119-SL1/28-ST1A	CANNA 147.1	119.3	27.8	147.1	30	7	2.25	2.25	6.75	15.80	547.4	54.03	0.2423	75500	1.80E-05	S
147-SL1/34-ST1A	CANNA 181.6	147.3	34.4	181.6	30	7	2.50	2.50	7.50	17.50	675.8	64.94	0.1963	75500	1.80E-05	S
185-SL1/43-ST1A	CANNA 228	184.7	43.1	227.8	30	7	2.80	2.80	8.40	19.60	847.7	80.54	0.1565	75500	1.80E-05	S
234-SL1/55-ST1A	CANNA 288	233.8	54.6	288.3	30	7	3.15	3.15	9.45	22.10	1072.8	98.58	0.1236	75500	1.80E-05	S

ALL ALUMINIUM ALLOY CONDUCTORS

AS PER EN 50182:2001 - CHARACTERISTICS OF ALUMINIUM ALLOY CONDUCTORS USED IN FRANCE - TYPE AL4

Code	Old Code	Area	No. of Wires	Diameter		Mass per Unit Length	Rated Strength	DC resstance	Final modulus of Elasticity	Coefficient of linera expansion	Direction of lay of external layer
				Wire	Cond.						
		mm^2		mm	mm	Kg/Km	kN	Ω/km	N/mm^2	1/K	
	ASTER										
22-AL4	22	22.0	7	2.00	6.00	60.0	7.15	1.4989	62000	2.30E-05	S
34-AL4	34.4	34.4	7	2.50	7.5	93.8	11.17	0.9593	62000	2.30E-05	S
55-AL4	54.6	54.6	7	3.15	9.45	148.9	17.73	0.6042	62000	2.30E-05	S
76-AL4	75.5	75.5	19	2.25	11.3	207.4	24.55	0.4388	62000	2.30E-05	S
117-AL4	117	117	19	2.80	14	321.2	38.02	0.2833	62000	2.30E-05	S
148-AL4	148	148.1	19	3.15	15.8	406.5	48.12	0.2239	62000	2.30E-05	S
182-AL4	181.6	181.6	37	2.50	17.5	500.3	59.03	0.1831	57000	2.30E-05	S
228-AL4	228	227.8	37	2.80	19.6	627.6	74.04	0.1460	57000	2.30E-05	S
288-AL4	288	288.3	37	3.15	22.1	794.3	93.71	0.1154	57000	2.30E-05	S
366-AL4	366	366.2	37	3.55	24.9	1008.9	115.36	0.0908	57000	2.30E-05	S
570-AL4	570	570.2	61	3.45	31.1	1576.0	185.33	0.0585	54000	2.30E-05	S
851-AL4	851	850.7	91	3.45	38.0	2360.7	276.47	0.0394	52500	2.30E-05	S
1144-AL4	1144	1143.5	91	4.00	44.0	3173.4	360.22	0.0293	52500	2.30E-05	S
1596-AL4	1600	1595.9	127	4.00	52.0	4427.5	502.72	0.0210	52500	2.30E-05	S

ALL ALUMINIUM CONDUCTORS (AAC)

AS PER IEC 61089 - 1991 (TYPE A1)

Code Number	Area	Number of Wires	Approx. Diameter		Approx. Weight	Approx. Calculated Breaking Load	Calculated Resistance at 20°c (Max.)
			Wire	Conductor			
	mm		mm	mm	kg / km	kn	ohm/km
10	10	7	1.35	4.05	27.4	1.95	2.8633
16	16	7	1.71	5.12	43.8	3.04	1.7896
25	25	7	2.13	6.40	68.4	4.50	1.1453
40	40	7	2.70	8.09	109.4	6.80	0.7158
63	63	7	3.39	10.20	172.3	10.39	0.4545
100	100	19	2.59	12.90	274.8	17.00	0.2877
125	125	19	2.89	14.50	343.6	21.25	0.2302
160	160	19	3.27	16.40	439.8	26.40	0.1798
200	200	19	3.66	18.30	549.7	32.00	0.1439
250	250	19	4.09	20.50	687.1	40.00	0.1151
315	315	37	3.29	23.00	867.9	51.97	0.0916
400	400	37	3.71	26.00	1102.0	64.00	0.0721
450	450	37	3.94	27.50	1239.8	72.00	0.0641
500	500	37	4.15	29.00	1377.6	80.00	0.0577
560	560	37	4.39	30.70	1542.9	89.60	0.0515
630	630	61	3.63	32.60	1739.3	100.80	0.0458
710	710	61	3.85	34.60	1959.1	113.60	0.0407
800	800	61	4.09	36.80	2207.4	128.00	0.0361
900	900	61	4.33	39.00	2488.3	144.00	0.0321
1000	1000	61	4.57	41.10	2759.2	160.00	0.0289
1120	1120	91	3.96	43.50	3093.5	179.20	0.0258
1250	1250	91	4.18	46.00	3452.6	200.00	0.0231
1400	1400	91	4.43	48.70	3866.9	224.00	0.0207
1500	1500	91	4.58	50.40	4143.1	240.00	0.0193

ALUMINIUM ALLOY CONDUCTORS STEEL REINFORCED (AACSR)

AS PER IEC 61089 - 1991

Code Number	Steel Ratio	Areas			No. of Wires		Wire Diameter		Diameter		Linear Mass	Rated Strength	D.C. Resistance
		Alum.	Steel	Total	Al	Steel	Al	Steel	Core	Cond.			
	%	mm^2	mm^2	mm^2			mm	mm	mm	Mm	kg/km	kn	ohm/km
16	17	18.4	3.07	21.5	6	1	1.98	1.98	1.98	5.93	74.4	9.02	1.7934
25	17	28.8	4.80	33.6	6	1	2.47	2.47	2.47	7.41	116.2	13.96	1.1478
40	17	46.0	7.67	53.7	6	1	3.13	3.13	3.13	9.38	185.9	22.02	0.7174
63	17	72.5	12.1	84.6	6	1	3.92	3.92	3.92	11.8	292.8	34.68	0.4555
100	6	115	6.39	121	18	1	2.85	2.85	2.85	14.3	366.4	41.24	0.2880
125	6	144	7.99	152	18	1	3.19	3.19	3.19	16.0	458.0	51.23	0.2304
125	16	144	23.4	167	26	7	2.65	2.06	6.19	16.8	579.9	69.86	0.2310
160	6	184	10.2	194	18	1	3.61	3.61	3.61	18.0	586.2	65.58	0.1800
160	16	184	30.0	214	26	7	3.00	2.34	7.01	19.0	742.3	88.52	0.1805
200	6	230	12.8	243	18	1	4.04	4.04	4.04	20.2	732.8	81.97	0.1440
200	16	230	37.5	268	26	7	3.36	2.61	7.83	21.3	927.9	110.64	0.1444
250	10	288	28.3	316	22	1	4.08	2.27	6.80	23.1	1013.5	117.09	0.1154
250	16	288	46.9	335	26	7	3.75	2.92	8.76	23.8	1159.8	138.31	0.1155
315	7	363	25.1	388	45	7	3.20	2.14	6.41	25.6	1196.5	136.28	0.0917
315	16	363	59.0	422	26	7	4.21	3.28	9.83	26.7	1461.4	171.90	0.0917
400	7	460	31.8	492	45	7	3.61	2.41	7.22	28.9	1519.4	172.10	0.0722
400	13	460	59.7	520	54	7	3.29	3.29	9.88	29.7	1738.3	201.46	0.0723
450	7	518	35.8	554	45	7	3.83	2.55	7.66	30.6	1709.3	193.61	0.0642
450	13	518	67.1	585	54	7	3.49	3.49	10.5	31.5	1955.6	226.64	0.0643
500	7	575	39.8	615	45	7	4.04	2.69	8.07	32.3	1899.3	215.12	0.0578
500	13	575	74.6	650	54	7	3.68	3.68	11.1	33.2	2172.9	251.82	0.0578
560	7	645	44.6	689	45	7	4.27	2.85	8.54	34.2	2127.2	240.93	0.0516
560	13	645	81.6	726	54	19	3.90	2.34	11.7	35.1	2420.9	283.21	0.0516
630	4	725	31.3	756	72	7	3.58	2.39	7.16	35.8	2248.0	249.62	0.0459
630	13	725	91.8	817	54	19	4.13	2.48	12.4	37.2	2723.5	318.61	0.0459
710	4	817	35.3	852	72	7	3.80	2.53	7.60	38.0	2533.4	281.32	0.0407
710	13	817	104	921	54	19	4.39	2.63	13.2	39.5	3069.4	359.06	0.0407
800	4	921	39.8	961	72	7	4.04	2.69	8.07	40.4	2854.6	316.98	0.0361
800	8	921	76.7	997	84	7	3.74	3.74	11.2	41.1	3145.1	356.03	0.0362
900	4	1036	44.8	1081	72	7	4.28	2.85	8.56	42.8	3211.4	356.60	0.0321
900	8	1036	86.3	1122	84	7	3.96	3.96	11.9	43.6	3538.3	400.53	0.0322
1000	8	1151	93.7	1245	84	19	4.18	2.51	12.5	45.9	3916.8	446.37	0.0289
1120	8	1289	105	1394	84	19	4.42	2.65	13.3	48.6	4386.8	499.93	0.0258

ALL ALUMINIUM ALLOY CONDUCTOR (AAAC)

AS PER IS -398-IV - 1994

Actual Area Sq. mm	tanding and Wire dia Nos./ mm	pprox. Overall dia mm	Approx . Kg / Km	Calculated Resistance Breaking Load Ohm/Km	Approx. Calculated at 20°C (Max) KN
15	3/2.50	5.39	40.15	2.3040	4.33
22	7/2.00	6.00	60.15	1.5410	6.45
34	7/2.50	7.50	94.00	0.9900	10.11
55	7/3.15	9.45	149.20	0.6210	16.03
80	7/3.81	11.43	218.26	0.4250	23.41
100	7/4.26	12.78	272.86	0.3390	26.26
125	19/2.89	14.45	342.51	0.2735	36.64
148	19/3.15	15.75	406.91	0.2290	43.50
173	19/3.40	17.00	474.02	0.1969	50.54
200	19/3.66	18.30	549.40	0.1710	58.66
232	19/3.94	19.70	636.67	0.1471	68.05
288	37/3.15	22.05	794.05	0.1182	84.71
346	37/3.45	24.15	952.56	0.0984	101.58
400	37/3.71	25.97	1101.63	0.0829	117.40
465	37/4.00	28.00	1280.50	0.0734	136.38
525	61/3.31	29.79	1448.39	0.0651	146.03
570	61/3.45	31.05	1573.71	0.0598	158.66
604	61/3.55	31.95	1666.00	0.0568	167.99
642	61/3.66	32.94	1771.36	0.0534	178.43
695	61/3.81	34.29	1919.13	0.0492	193.25
767	61/4.00	36.00	2115.54	0.0446	213.01

AS PER BS EN 50182 TYPE-AL3

Code Word	Sectional Area	Stranding		Diameter of Complete Conductor	Weight	Rated Strength	DC Resistance @ 20°C	Current Capacity	
		No. of Aluminium Wires	Individual wire diameter					@ 75°C	@ 85°C
	(mm^2)	(No.)	(mm)	(mm)	(Kg/Km)	KN	(Ω/Km)	(Ampere)	(Ampere)
BOX	18.8	7	1.85	5.55	51.4	5.55	1.748	84	99
ACACIA	23.8	7	2.08	6.24	64.9	7.02	1.3828	97	115
ALMOND	30.1	7	2.34	7.02	82.2	8.88	1.0926	111	133
CEDAR	35.5	7	2.54	7.62	96.8	10.46	0.9273	123	147
DEODAR	42.2	7	2.77	8.31	115.2	12.44	0.7797	136	163
FIR	47.8	7	2.95	8.85	130.6	14.11	0.6875	147	176
HAZEL	59.9	7	3.3	9.9	163.4	17.66	0.5494	167	202
PINE	71.6	7	3.61	10.8	195.6	21.14	0.4591	186	225
HOLLY	84.1	7	3.91	11.7	229.5	24.79	0.3913	205	248
WILLOW	89.7	7	4.04	12.1	245	26.47	0.3665	213	258
OAK	118.9	7	4.65	14	324.5	35.07	0.2767	251	306
MULBERRY	150.9	19	3.18	15.9	414.3	44.52	0.2192	288	354
ASH	180.7	19	3.48	17.4	496.1	53.31	0.183	320	394
ELM	211	19	3.76	18.8	579.2	62.24	0.1568	350	433
POPLAR	239.4	37	2.87	20.1	659.4	70.61	0.1387	376	467
SYCAMORE	303.2	37	3.23	22.6	835.2	89.4	0.1095	430	538
UPAS	362.1	37	3.53	24.7	997.5	106.82	0.0917	475	598
YEW	479	37	4.06	28.4	1319.6	141.31	0.0693	555	706
TOTARA	498.1	37	4.14	29	1372.1	146.93	0.0666	567	722
RUBUS	586.9	61	3.5	31.5	1622	173.13	0.0567	620	794
SORBUS	659.4	61	3.71	33.4	1822.5	194.53	0.0505	659	849
ARAUCARIA	821.1	61	4.14	37.3	2269.4	242.24	0.0406	740	962
REDWOOD	996.2	61	4.56	41	2753.2	293.88	0.0334	816	1072

AS PER AMERICAN SPECIFICATIONS ASTM B-399 - 2016

Nominal Alunium Sq. mm	tranding & Wire Dia No. / mm	Approx. Overall Diameter mm	Sectional Area Sq. mm	Approx Weight Kg/Km	pprox Calculated Breaking Load KN	Calculated Resistance at 20 °C (Max) Ohm/Km
16	7/1.70	5.10	15.89	43	4.44	2.0910
25	7/2.10	6.30	24.25	66	6.77	1.3720
35	7/2.50	7.50	34.36	94	9.6	0.9670
50	7/3.00	9.00	49.48	135	13.82	0.6716
50	19/1.80	9.00	48.35	133	13.5	0.6893
70	19/2.10	10.50	65.81	181	18.38	0.5079
95	19/2.50	12.50	93.27	256	26.05	0.3580
120	19/2.80	14.00	117.00	322	32.68	0.2853
150	37/2.25	15.00	147.11	406	41.09	0.2278
185	37/2.50	17.50	181.62	500	50.73	0.1842
240	61/2.25	20.30	242.54	670	67.74	0.1385
300	61/2.50	22.50	299.43	827	83.63	0.1119
400	61/2.89	26.00	400.14	1104	111.76	0.0841
500	61/3.23	29.10	499.83	1379	139.6	0.0671
625	91/2.96	32.60	626.20	1720	174.9	0.0538
800	91/3.35	36.90	802.09	2218	224.02	0.0420
1000	91/3.74	41.10	999.71	2767	279.22	0.0336

ALUMINUM ALLOY CONDUCTORS (TYPE- AL59) SS 424 08 14

Nominal Area	Sectional Area	Equivalent copper area	Stranding		Diameter of Complete Conductor	Weight	Rated Strength	DC Resistance @ 20°C	Current Capacity	
			No. of Aluminium Wires	Individual wire diameter					@ 75°C	@ 85°C
(mm^2)	(mm^2)	(mm^2)	(No.)	(mm)	(mm)	(Kg/Km)	KN	(Ω/Km)	(Ampere)	(Ampere)
31	31.14	18.48	7	2.38	7.14	85	7.77	0.943	120	142
62	62.44	37.06	7	3.37	10.11	170	15.6	0.47	181	218
99	99.3	58.93	7	4.25	12.80	271	22.8	0.296	238	289
157	158.59	94.12	19	3.26	16.3	436	39.7	0.186	313	384
241	241.16	143.13	19	4.02	20.1	663	55.5	0.123	397	493
329	330.03	195.87	37	3.37	23.6	910	82.5	0.0899	475	595
454	454.49	269.74	61	3.08	27.72	1260	113	0.0654	567	719
593	593.62	352.31	61	3.52	31.68	1640	143	0.0501	655	840
774	774.24	459.51	61	4.02	36.18	2140	178	0.0384	754	977
910	910.74	540.52	61	4.36	39.2	2520	209	0.0326	819	1070

ECO CONDUCTORS

Eco Series conductors consists of trapezoidal shaped special Aluminium Alloy (AL-59) containing Magnesium, Silicon and Copper stranded with round core. The trapezoidal shape provides up to 96% compactness.

REF. SPECIFICATION - SS 42408013, 14

Code Word	Size	Appropriate System Voltage	Equivalent ACSR	Reference Standards	Conductor diameter	Weight	Ultimate tensile strength	DC resistance at 20°C temperature	Current carrying capacity at maximum operating temperature	Power Transfer Capacity (at 33 kV at 0.95 PF)
	Sq. mm.	kV			mm	kg/km	kN	Ohm/km	A	MW/CKT
ECO DOG	139	S110 kV	DOG	SS 42401813, SS 42401814 & IEC 62219 & IS 398 Part IV	14.15	380	31.0	0.2181	393	21
ECO PANTHER	310	S132 kV	PANTHER		21.2	855	64.8	0.0987	641	34
ECO PANTHER	353	S132 kV	PANTHER		22	973	81.95	0.08	698	37

GALVANIZED STEEL WIRE EARTHWIRE / STAYWIRE / GUYWIRE / SPACER CABLE (GSW)

REF. SPECIFICATION -IS 2141

Sectional Area	Stranding		Diameter of EARTHWIRE	Weight	Rated Strength				
	No. of Wires	Individual wire diameter			Grade 1770	Grade 1570	Grade 1100	Grade 700	'Grade 450
mm²	No.	mm	mm	Kg/km	KN	KN	KN	KN	KN
23.38	3	3.15	6.79	-	40.10	35.60	24.90	15.90	10.20
31.17	4	3.15	7.60	-	52.40	46.50	32.60	20.70	13.30
2.01	4	0.80	2.40	-	5.50	4.90	3.40	2.20	1.40
4.91	4	1.25	3.75	-	13.50	12.00	8.40	5.40	3.40
3.14	4	1.00	4.80	-	22.20	19.70	13.80	8.80	5.60
12.57	4	2.00	6.00	-	34.70	30.80	21.60	13.70	8.80
27.59	7	2.24	6.72	-	43.50	38.60	27.00	17.20	11.10
34.36	7	2.50	7.50	-	54.20	48.10	33.70	21.40	13.80
54.55	7	3.15	9.45	-	86.00	76.30	53.50	34.00	21.90
87.96	7	4.00	12.00	-	139.00	123.00	86.20	54.90	35.30
3.52	7	0.80	4.00	-	14.90	13.20	9.20	5.90	3.80
8.59	7	1.25	6.25	-	36.30	32.20	22.60	14.40	9.20
14.07	7	1.60	8.00	-	59.50	52.80	37.00	23.50	15.10
59.69	19	2.00	10.00	-	93.00	82.50	57.80	36.80	23.60
74.88	19	2.24	11.20	-	117.00	103.00	72.50	46.10	29.70
93.27	19	2.50	12.50	-	145.00	129.00	90.30	57.50	36.90
238.76	19	4.00	20.00	-	372.00	330.00	231.00	147.00	94.60
9.55	19	0.80	5.60	-	28.80	25.50	17.90	11.40	7.30
23.32	19	1.25	8.75	-	70.30	62.30	43.70	27.80	17.90
38.20	19	1.60	11.20	-	115.00	102.00	71.50	45.50	29.30
148	19	3.15	15.75	-	231.00	205.00	143.00	91.20	58.60

REF. SPECIFICATION - IS 12776

Sectional Area	Stranding		Diameter of EARTHWIRE	Weight	Rated Strength		
	No. of Wires	Individual wire diameter			Grade 981	Grade 1100	Grade 1310
mm²	No.	mm	mm	Kg/km	KN	KN	KN
49.48	7	3.00	9.00	386.00	43.70	49.00	58.30
54.55	7	3.15	9.45	426.00	48.20	54.00	64.30
58.07	7	3.25	9.75	454.00	51.30	57.50	68.50
60.96	7	3.33	9.99	476.00	53.80	60.40	71.90
73.65	7	3.66	10.98	575.00	65.00	72.90	86.80
87.96	7	4.00	12.00	687.00	77.70	87.10	104.00

REF. SPECIFICATION - ASTM A 363

Sectional Area	Stranding		Diameter of EARTHWIRE	Weight	Rated Strength		
	No. of Wires	Individual wire diameter			High Strength Grade	Extra High Strength Grade	UTILITY GRADE
mm²	No.	mm	mm	Kg/km	KN	KN	KN
31.91	3	3.68	7.94	255.00	-	-	28.90
38.32	7	2.64	7.94	305.00	35.60	49.8	-
51.14	7	3.05	9.52	407.00	48.00	68.4	51.20
74.45	7	3.68	11.11	595.00	64.50	92.5	-
96.52	7	4.19	12.70	770.00	83.60	119.6	-

GALVANIZED STEEL WIRE EARTHWIRE / STAYWIRE / GUYWIRE / SPACER CABLE (GSW)

REF. SPECIFICATION -ASTM A 475

Sectional Area	Stranding		Diameter of EARTHWIRE	Weight	Rated Strength				
	No. of Wires	Individual wire diameter			Common Grade	Siemens Martin Grade	High Strength Grade	Extra High Strength Grade	UTILITY GRADE
mm²	No.	mm	mm	Kg/km	KN	KN	KN	KN	KN
5.95	7	1.04	3.18	48.00	2.402	4.048	5.916	8.140	-
9.58	7	1.32	3.97	76.00	3.870	6.539	6.539	13.078	-
13.55	7	1.57	4.76	109.00	5.115	8.452	12.677	17.748	-
14.97	7	1.65	4.76	119.00	-	-	-	-	10.676
16.42	3	2.64	5.56	131.00	6.228	10.409	15.569	21.796	-
18.41	7	1.83	5.56	146.00	6.850	11.387	17.126	24.020	-
21.92	3	3.05	6.35	174.00	8.274	13.523	21.040	29.981	14.01
21.92	3	3.05	6.35	174.00	-	-	-	-	20.02
22.66	7	2.03	6.35	180.00	8.452	14.012	21.129	29.581	-
25.66	3	3.30	7.14	204.00	9.252	15.035	23.398	33.362	-
30.62	7	2.36	7.14	244.00	11.432	18.905	28.469	39.812	20.46
31.91	3	3.68	7.94	255.00	11.076	18.193	28.246	40.479	26.91
38.32	7	2.64	7.94	305.00	14.234	23.798	35.586	49.820	-
42.18	7	2.77	7.94	335.00	-	-	-	-	26.69
41.37	3	4.19	9.52	328.00	14.813	24.732	37.187	52.489	37.81
51.14	7	3.05	9.52	407.00	18.905	30.915	48.040	68.503	51.16
74.45	7	3.68	11.11	595.00	25.355	41.591	64.499	92.523	80.07
96.52	7	4.19	12.70	770.00	32.917	53.823	83.627	119.657	111.21
96.27	19	2.54	12.70	751.00	33.895	56.492	84.961	118.768	-
125.62	7	4.78	14.29	1000.00	42.703	69.837	108.981	155.688	-
122.92	19	2.87	14.29	949.00	42.881	71.616	107.202	149.905	-
152.11	7	5.26	15.88	1211.00	51.599	84.961	131.667	188.605	-
150.90	19	3.18	15.88	1186.00	48.930	80.513	124.995	178.819	-
216.62	19	3.81	19.05	1721.00	71.172	116.543	181.487	259.331	-
302.18	19	4.50	22.22	2356.00	97.416	159.691	248.211	354.523	-
385.10	19	5.08	25.40	3089.00	127.664	209.07	325.61	464.8	-

REF. SPECIFICATION -DIN 48201

Nominal Area	Sectional Area	Stranding		Diameter of EARTHWIRE	Weight	Rated Strength			
		No. of Wires	Individual wire diameter			ST I	ST II	ST III	ST IV
mm²	mm²	No.	mm	mm	Kg/km	KN	KN	KN	KN
25	24.25	7	2.10	6.30	192.00	8.98	15.90	30.18	36.17
35	34.36	7	2.50	7.50	272.00	12.73	22.52	42.76	51.25
50	49.48	7	3.00	9.00	391.00	18.33	32.43	61.58	73.80
50	48.35	19	1.80	9.00	384.00	17.91	31.69	60.17	72.11
70	65.81	19	2.10	10.50	522.00	24.38	43.14	81.90	98.16
95	93.27	19	2.50	12.50	741.00	34.56	61.14	116.07	139.11
120	116.99	19	2.80	14.00	929.00	43.34	76.69	145.59	174.49

TECHNICAL DETAILS FOR 1.1 KV SINGLE CORE, ALUMINIUM / COPPER CONDUCTOR, PVC INSULATED, UN-ARMOURED CABLES

PHYSICICAL PARAMETER :: CABLE CODE - AYY/YY, REF SPECIFICATION : IS 1554 PART-1

Size (Cross Section al Area)	Minium No. of Strand in Conductor		Nominal Insulation Thickness	Nominal Sheath Thickness	Approx. Overall Dia of Cable	Approx. Weight of Cable	
						With Al Conductor	With Cu Conductor
	Aluminium	Copper				AYY	YY
Sq. mm	Nos	Nos	mm	mm	mm	Kg/Km	Kg/Km
4	1/3	1/7	1.00	1.80	8	80	100
6	1/3	1/7	1.00	1.80	9	90	120
10	1/7	1/7/6	1.00	1.80	10	110	170
16	6	6	1.00	1.80	10	150	250
25	6	6	1.20	1.80	12	200	350
35	6	6	1.20	1.80	13	250	450
50	6	6	1.40	1.80	15	300	600
70	12	12	1.40	1.80	16	400	800
95	15	15	1.60	1.80	18	500	1050
120	15	18	1.60	2.00	20	600	1300
150	15	18	1.80	2.00	22	700	1600
185	30	30	2.00	2.00	24	850	1950
240	30	34	2.20	2.00	27	1100	2500
300	30	34	2.40	2.00	30	1300	3100
400	53	53	2.60	2.20	34	1700	3950
500	53	53	3.00	2.20	37	2100	5000
630	53	53	3.40	3.40	42	2700	6450
800	53	53	3.40	2.40	46	3250	8050
1000	53	53	3.40	2.60	50	3950	9950

* Tabulated Approx. Weight of Cable are only for the purpose of guidline for transportation, loading & unloading purpose.

Cross-sectional view

1. Conductor Material: Aluminium / Copper asper Class-2 of IS 8130
   Shape of Conductor: For Aluminium Conductor - 4Sq.mm, 6Sq.mm & 10 Sq. mm Solid/ Stranded Non Compacted Circular, 16 Sq. mm and above - Stranded Compacted Circular
   For Copper Conductor - 4 Sq. mm & 6 Sq. mm Solid / Stranded Non Compacted Circular, 10 Sq. mm and above - Stranded Compacted Circular
2. Insulation Material: PVC Type-A or HR PVC Type -C as per IS 5831; Colour : Black, Red
3. Outer Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS type, LSOH
   Colour of Outer Sheath: black or any other Colour as per requirement

ELECTRICAL PARAMETER

Size (Cross Sectional Area)	Max. Conductor D.C. Resistance at 20 °C		Approx. Conductor A.C. Resistance at 70 °C		Reactance of Cable at 50Hz (Approx)	Capacitance of Cable (Approx)	Normal Current Rating						Short Circuit Current Rating for 1s Duration	
							For Aluminium Conductor			For Aluminium Conductor				
	Aluminium	Copper	Aluminium	Copper			Ground	Duct	Air	Ground	Duct	Air	Aluminium	Copper
Sq. mm	Ω/Km	Ω/Km	Ω/Km	Ω/Km	Ω/Km	µF/Km	A	A		A	A	A	kA	kA
4	7.4100	4.6100	8.8900	5.5300	0.140	0.580	36	33	32	46	42	43	0.304	0.460
6	4.6100	3.0800	5.5300	3.7000	0.127	0.680	44	42	41	57	54	54	0.456	0.690
10	3.0800	1.8300	3.7000	2.2000	0.118	0.830	59	56	56	75	72	72	0.760	1.150
16	1.9100	1.1500	2.2900	1.3800	0.110	1.010	75	71	72	94	92	92	1.220	1.840
25	1.2000	0.7270	1.4400	0.8700	0.105	1.050	97	93	99	125	120	125	1.900	2.880
35	0.8680	0.5240	1.0400	0.6300	0.100	1.220	120	110	120	150	140	155	2.660	4.030
50	0.6410	0.3870	0.7690	0.4640	0.098	1.220	145	130	150	180	165	190	3.800	5.750
70	0.4430	0.2680	0.5320	0.3220	0.091	1.430	170	155	185	220	200	235	5.320	8.050
95	0.3200	0.1930	0.3840	0.2320	0.088	1.470	205	180	215	265	230	275	7.220	10.900
120	0.2530	0.1530	0.3040	0.1840	0.086	1.620	230	200	240	300	255	310	9.120	13.800
150	0.2060	0.1240	0.2470	0.1488	0.085	1.620	265	220	270	340	280	345	11.400	17.300
185	0.1640	0.0991	0.1970	0.1189	0.084	1.620	300	240	305	380	305	390	14.100	21.300
240	0.1250	0.0754	0.1510	0.0912	0.082	1.720	335	270	350	420	340	445	18.200	27.300
300	0.1000	0.0601	0.1220	0.0733	0.080	1.740	370	295	395	465	370	500	22.800	34.500
400	0.0778	0.0470	0.0961	0.0580	0.080	1.810	410	335	455	500	405	510	30.400	46.000
500	0.0605	0.0366	0.0759	0.0459	0.079	1.860	435	355	490	540	430	610	38.000	57.500
630	0.0469	0.0283	0.0610	0.0368	0.077	1.870	485	395	560	590	465	680	47.900	72.500
800	0.0367	0.0221	0.0503	0.0303	0.077	1.980	525	420	650	645	500	745	60.800	92.000
1000	0.0291	0.0176	0.0422	0.0255	0.076	2.200	570	445	735	705	546	890	76.000	115.000

Note: Normal Current ratings are given in standard conditions, if site conditions are diffferent , current rating should be multipled by rating factor.

TECHNICAL DETAILS FOR 1.1 KV TWO CORE, ALUMINIUM / COPPER CONDUCTOR, PVC INSULATED, UN-ARMOURED CABLES

PHYSICICAL PARAMETER :: CABLE CODE - AYY/YY, REF SPECIFICATION : IS 1554 PART-1

Size (Cross Section al Area)	Minium No. of Strand in Conductor		Nominal Insulation Thickness	Minimum Inner Sheath Thickness	Nominal Outer Sheath Thickness	Approx. Overall Dia of Cable	Approx. Weight of Cable	
							With Al Conductor	With Cu Conductor
	Aluminium	Copper					AYY	YY
Sq. mm	Nos	Nos	mm	mm	mm	mm	Kg/Km	Kg/Km
4	1/3	1/7	1.00	0.3	1.80	13	230	280
6	1/3	1/7	1.00	0.3	1.80	15	270	350
10	1/7	1/7/6	1.00	0.3	1.80	16	330	480
16	6	6	1.00	0.3	1.80	16	350	500
25	6	6	1.20	0.3	2.00	19	450	750
35	6	6	1.20	0.3	2.00	20	550	950
50	6	6	1.40	0.3	2.00	23	700	1250
70	12	12	1.40	0.3	2.00	25	850	1650
95	15	15	1.60	0.4	2.20	29	1150	2250
120	15	18	1.60	0.4	2.20	31	1300	2700
150	15	18	1.80	0.4	2.40	33	1600	3300
185	30	30	2.00	0.5	2.40	36	1900	4100
240	30	34	2.20	0.5	2.60	42	2450	5250
300	30	34	2.40	0.6	2.80	45	2950	6500
400	53	53	2.60	0.7	3.20	51	3800	8300
500	53	53	3.00	0.7	3.40	57	4750	10550
630	53	53	3.40	0.7	3.80	64	6000	13500

* Tabulated Approx. Weight of Cable are only for the purpose of guidline for transportation, loading & unloading purpose.

Cross-sectional view

1. Conductor Material: Aluminium / Copper as per Class-2 of IS 8130
   Shape of Conductor: For Aluminium Conductor - 4 Sq.mm, 6 Sq.mm & 10 Sq. mm Solid/ Stranded Non Compacted Circular, 16 Sq. mm and above - Stranded Compacted for Copper Conductor - 4 Sq. mm & 6 Sq. mm Solid / Stranded Non Compacted Circular, 10 Sq. mm and above - Stranded Compacted Circular, 16 Sq.mm and above - Stranded Compacted Shaped
2. Insulation Material: PVC Type-A or HR PVC Type -C as per IS 5831; Colour : Red & Black
3. Inner Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS Type, LSOH
4. Outer Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS type, LSOH
   Colour of Outer Sheath: Black or any other Colour as per requirement

ELECTRICAL PARAMETER

Size (Cross Sectional Area)	Max. Conductor D.C. Resistance at 20 ^{0}C		Approx. Conductor A.C. Resistance at 70 ^{0}C		Reactance of Cable at 50 Hz (Approx)	Capacitance of Cable (Approx)	Normal Current Rating						Short Circuit Current Rating for 1s Duration	
							For Aluminium Conductor			For Aluminium Conductor				
	Aluminium	Copper	Aluminium	Copper			Ground	Duct	Air	Ground	Duct	Air	Aluminium	Copper
Sq. mm	Ω/Km	Ω/Km	Ω/Km	Ω/Km	Ω/Km	µF/Km	A	A		A	A	A	kA	kA
4	7.4100	4.6100	8.8900	5.5300	0.098	0.23	32	27	27	41	35	35	0.304	0.460
6	4.6100	3.0800	5.5300	3.7000	0.096	0.28	40	34	35	50	44	45	0.456	0.690
10	3.0800	1.8300	3.7000	2.2000	0.091	0.34	55	45	47	70	58	60	0.760	1.150
16	1.9100	1.1500	2.2900	1.3800	0.085	0.40	70	58	59	90	75	78	1.220	1.840
25	1.2000	0.7270	1.4400	0.8700	0.083	0.42	90	76	48	115	97	105	1.900	2.880
35	0.8680	0.5240	1.0400	0.6300	0.082	0.48	110	92	99	140	125	125	2.660	4.030
50	0.6410	0.3870	0.7690	0.4640	0.082	0.49	135	115	125	165	145	155	3.800	5.750
70	0.4430	0.2680	0.5320	0.3220	0.076	0.56	160	140	150	205	180	195	5.320	8.050
95	0.3200	0.1930	0.3840	0.2320	0.076	0.58	190	170	185	240	215	230	7.220	10.900
120	0.2530	0.1530	0.3040	0.1840	0.075	0.63	210	190	210	275	235	265	9.120	13.800
150	0.2060	0.1240	0.2470	0.1488	0.074	0.63	240	210	240	310	270	305	11.400	17.300
185	0.1640	0.0991	0.1970	0.1189	0.074	0.64	275	240	275	350	300	350	14.100	21.300
240	0.1250	0.0754	0.1510	0.0912	0.073	0.67	320	275	325	405	345	410	18.200	27.300
300	0.1000	0.0601	0.1220	0.0733	0.073	0.68	355	305	365	450	385	465	22.800	34.500
400	0.0778	0.0470	0.0961	0.0580	0.072	0.70	385	345	420	490	425	530	30.400	46.000
500	0.0605	0.0366	0.0759	0.0459	0.072	0.70	425	380	475	540	460	605	38.000	57.500
630	0.0469	0.0283	0.0610	0.0368	0.072	0.70	465	415	540	640	550	785	47.900	72.500

Note: Normal Current ratings are given in standard conditions, if site conditions are diffferent , current rating should be multipled by rating factor.

TECHNICAL DETAILS FOR 1.1 KV THREE CORE, ALUMINIUM / COPPER CONDUCTOR, PVC INSULATED, UN-ARMOURED CABLES

PHYSICICAL PARAMETER :: CABLE CODE - AYY/YY, REF SPECIFICATION : IS 1554 PART-1

Size (Cross Section al Area)	Minium No. of Strand in Conductor		Nominal Insulation Thickness	Minimum Inner Sheath Thickness	Nominal Outer Sheath Thickness	Approx. Overall Dia of Cable	Approx. Weight of Cable	
							With Al Conductor	With Cu Conductor
	Aluminium	Copper					AYY	YY
Sq. mm	Nos	Nos	mm	mm	mm	mm	Kg/Km	Kg/Km
4	1/3	1/7	1.00	0.3	1.80	14	250	330
6	1/3	1/7	1.00	0.3	1.80	15	310	420
10	1/7	7/6	1.00	0.3	1.80	17	370	580
16	6	6	1.00	0.3	1.80	18	450	700
25	6	6	1.20	0.3	2.00	21	600	1050
35	6	6	1.20	0.3	2.00	23	700	1350
50	6	6	1.40	0.3	2.00	26	900	1750
70	12	12	1.40	0.3	2.00	29	1200	2400
95	15	15	1.60	0.4	2.20	33	1500	3200
120	15	18	1.60	0.4	2.20	36	1800	3900
150	15	18	1.80	0.4	2.40	40	2200	4800
185	30	30	2.00	0.5	2.40	43	2700	5950
240	30	34	2.20	0.5	2.80	49	3550	7750
300	30	34	2.40	0.6	3.00	54	4200	9600
400	53	53	2.60	0.7	3.40	61	5350	12200
500	53	53	3.00	0.7	3.60	69	6750	15500
630	53	53	3.40	0.7	4.00	77	8550	19900

* Tabulated Approx. Weight of Cable are only for the purpose of guidline for transportation, loading & unloading purpose.

Cross-sectional view

1. Conductor Material: Aluminium / Copper as per Class-2 of IS 8130
   Shape of Conductor: For Aluminium Conductor - 4 Sq.mm, 6 Sq.mm & 10 Sq. mm Solid/ Stranded Non Compacted Circular,
   16 Sq. mm and above - Stranded Compacted shaped for Copper Conductor - 4 Sq. mm & 6 Sq. mm Solid / Stranded Non Compacted Circular, 10 Sq. mm
   - Stranded Compacted Circular, 16 Sq.mm and above - Stranded Compacted Shaped
2. Insulation Material: PVC Type-A or HR PVC Type -C as per IS 5831; Colour : Red, Yellow & Black
3. Inner Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS Type, LSOH
4. Outer Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS type, LSOH
   Colour of Outer Sheath: Black or any other Colour as per requirement

ELECTRICAL PARAMETER

Size (Cross Sectional Area)	Max. Conductor D.C. Resistance at 20 ^{0}C		Approx. Conductor A.C. Resistance at 70 ^{0}C		Reactance of Cable at 50 Hz (Approx)	Capacitance of Cable (Approx)	Normal Current Rating						Short Circuit Current Rating for 1s Duration	
							For Aluminium Conductor			For Aluminium Conductor				
	Aluminium	Copper	Aluminium	Copper			Ground	Duct	Air	Ground	Duct	Air	Aluminium	Copper
Sq. mm	Ω/Km	Ω/Km	Ω/Km	Ω/Km	Ω/Km	µF/Km	A	A		A	A	A	kA	kA
4	7.4100	4.6100	8.8900	5.5300	0.098	0.23	28	23	23	36	30	30	0.304	0.460
6	4.6100	3.0800	5.5300	3.7000	0.096	0.28	35	30	30	45	38	39	0.456	0.690
10	3.0800	1.8300	3.7000	2.2000	0.091	0.34	46	39	40	60	50	52	0.760	1.150
16	1.9100	1.1500	2.2900	1.3800	0.085	0.40	60	50	51	77	64	66	1.220	1.840
25	1.2000	0.7270	1.4400	0.8700	0.083	0.42	76	63	70	99	81	90	1.900	2.880
35	0.8680	0.5240	1.0400	0.6300	0.082	0.48	92	77	86	120	99	110	2.660	4.030
50	0.6410	0.3870	0.7690	0.4640	0.082	0.49	110	95	105	145	125	135	3.800	5.750
70	0.4430	0.2680	0.5320	0.3220	0.076	0.56	135	115	130	175	150	165	5.320	8.050
95	0.3200	0.1930	0.3840	0.2320	0.076	0.58	165	140	155	210	175	200	7.220	10.900
120	0.2530	0.1530	0.3040	0.1840	0.075	0.63	185	155	180	240	195	230	9.120	13.800
150	0.2060	0.1240	0.2470	0.1488	0.074	0.63	210	175	205	270	22	265	11.400	17.300
185	0.1640	0.0991	0.1970	0.1189	0.074	0.64	235	200	240	300	255	305	14.100	21.300
240	0.1250	0.0754	0.1510	0.0912	0.073	0.67	275	235	280	345	295	355	18.200	27.300
300	0.1000	0.0601	0.1220	0.0733	0.073	0.68	305	260	315	385	335	400	22.800	34.500
400	0.0778	0.0470	0.0961	0.0580	0.072	0.70	335	290	375	425	360	455	30.400	46.000
500	0.0605	0.0366	0.0759	0.0459	0.072	0.70	370	320	425	470	390	520	38.000	57.500
630	0.0469	0.0283	0.0610	0.0368	0.072	0.70	405	350	480	555	470	675	47.900	72.500

Note: Normal Current ratings are given in standard conditions, if site conditions are diffferent , current rating should be multipled by rating factor.

TECHNICAL DETAILS FOR 1.1 KV THREE & HALF CORE, ALUMINIUM / COPPER CONDUCTOR, PVC INSULATED, UN-ARMOURED CABLES

PHYSICICAL PARAMETER :: CABLE CODE - AYY/YY, REF SPECIFICATION : IS 1554 PART-1

Size (Cross Section al Area)	Minium No. of Strand in Conductor		Nominal Insulation Thickness	Minimum Inner Sheath Thickness	Nominal Outer Sheath Thickness	Approx. Overall Dia of Cable	Approx. Weight of Cable	
							With Al Conductor	With Cu Conductor
	Aluminium	Copper					AYY	YY
Sq. mm	Nos	Nos	mm	mm	mm	mm	Kg/Km	Kg/Km
3x25+16	6/6	6/6	1.20/1.00	0.3	2.00	23	700	1250
3x35+16	6/6	6/6	1.20/1.00	0.3	2.00	25	800	1550
3x50+25	6/6	6/6	1.40/1.20	0.3	2.20	28	1050	2050
3x70+35	12/6	12/6	1.40/1.20	0.4	2.20	32	1400	2800
3x95+50	15/6	15/6	1.60/1.40	0.4	2.20	36	1800	3700
3x120+70	15/12	18/12	1.60/1.40	0.5	2.40	39	2200	4700
3x150+70	15/12	18/12	1.80/1.40	0.5	2.40	43	2550	5550
3x185+95	30/15	30/15	2.00/1.60	0.5	2.60	47	3150	6900
3x240+120	30/15	34/18	2.20/1.60	0.6	3.00	53	4050	8950
3x300+150	30/15	34/18	2.40/1.80	0.6	3.20	58	4900	11100
3x400+185	53/30	53/30	2.60/2.00	0.7	3.40	64	6150	14000
3x500+240	53/30	53/34	3.00/2.20	0.7	3.80	76	7900	18050
3x630+300	53/30	53/34	3.40/2.40	0.7	4.00	84	9900	22950

* Tabulated Approx. Weight of Cable are only for the purpose of guidline for transportation, loading & unloading purpose.

Cross-sectional view

1. Conductor Material: Aluminium / Copper as per Class-2 of IS 8130
   Shape of Conductor: For Aluminium Conductor - 4 Sq.mm, 6 Sq.mm & 10 Sq. mm Solid/ Stranded Non Compacted Circular, 16 Sq. mm and above - Stranded Compacted Shaped for Copper Conductor - 4 Sq. mm & 6 Sq. mm Solid / Stranded Non Compacted Circular, 10 Sq. mm - Stranded Compacted Circular, 16 Sq.mm and above - Stranded Compacted Shaped
2. Insulation Material: PVC Type-A or HR PVC Type -C as per IS 5831; Colour : Red, Yellow, Blue & Black
3. Inner Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS Type, LSOH
4. Outer Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS type, LSOH
   Colour of Outer Sheath: Black or any other Colour as per requirement

ELECTRICAL PARAMETER

Size (Cross Sectional Area)	Max. Conductor D.C. Resistance at 20 ^{0}C		Approx. Conductor A.C. Resistance at 70 ^{0}C		Reactance of Cable at 50 Hz (Approx)	Capacitance of Cable (Approx)	Normal Current Rating						Short Circuit Current Rating for 1s Duration	
							For Aluminium Conductor			For Aluminium Conductor				
	Aluminium	Copper	Aluminium	Copper			Ground	Duct	Air	Ground	Duct	Air	Aluminium	Copper
Sq. mm	Ω/Km	Ω/Km	Ω/Km	Ω/Km	Ω/Km	µF/Km	A	A		A	A	A	kA	kA
3x25+16	1.2000	0.7270	1.4400	0.8700	0.083	0.42	76	63	70	99	81	90	1.900	2.880
3x35+16	0.8680	0.5240	1.0400	0.6300	0.082	0.48	92	77	86	120	99	110	2.660	4.030
3x50+25	0.6410	0.3870	0.7690	0.4640	0.082	0.49	11	95	105	145	125	135	3.800	5.750
3x70+35	0.4430	0.2680	0.5320	0.3220	0.076	0.56	135	115	130	175	150	165	5.320	8.050
3x95+50	0.3200	0.1930	0.3840	0.2320	0.076	0.58	165	140	155	210	175	200	7.220	10.900
3x120+70	0.2530	0.1530	0.3040	0.1840	0.075	0.63	185	155	180	240	195	230	9.120	13.800
3x150+70	0.2060	0.1240	0.2470	0.1488	0.074	0.63	210	175	205	270	225	265	11.400	17.300
3x185+95	0.1640	0.0991	0.1970	0.1189	0.074	0.64	235	200	240	300	255	305	14.100	21.300
3x240+120	0.1250	0.0754	0.1510	0.0912	0.073	0.67	275	235	280	345	295	355	18.200	27.600
3x300+150	0.1000	0.0601	0.1220	0.0733	0.073	0.68	305	260	315	385	335	400	22.800	34.500
3x400+185	0.0778	0.0470	0.0759	0.0580	0.072	0.70	335	290	375	425	350	455	30.400	46.000
3x500+240	0.0605	0.0366	0.0759	0.0459	0.072	0.70	370	320	425	470	390	520	38.000	57.500
3x630+300	0.0469	0.0283	0.0610	0.0368	0.072	0.70	405	350	480	555	470	675	47.900	72.500

Note: Normal Current ratings are given in standard conditions, if site conditions are diffferent , current rating should be multipled by rating factor.

TECHNICAL DETAILS FOR 1.1 KV THREE CORE, A LUMINIUM / COPPER CONDUCTOR, PVC INSULATED, UN-ARMOURED CABLES

PHYSICICAL PARAMETER :: CABLE CODE - AYY/YY, REF SPECIFICATION : IS 1554 PART-1

Size (Cross Section al Area)	Minium No. of Strand in Conductor		Nominal Insulation Thickness	Minimum Inner Sheath Thickness	Nominal Outer Sheath Thickness	Approx. Overall Dia of Cable	Approx. Weight of Cable	
							With Al Conductor	With Cu Conductor
	Aluminium	Copper					AYY	YY
Sq. mm	Nos	Nos	mm	mm	mm	mm	Kg/Km	Kg/Km
4	1/3	1/7	1.00	0.3	1.80	15	290	400
6	1/3	1/7	1.00	0.3	1.80	17	350	510
10	1/7	1/6/7	1.00	0.3	1.50	19	440	710
16	6	6	1.00	0.3	2.00	21	550	950
25	6	6	1.20	0.3	2.00	23	750	1350
35	6	6	1.20	0.3	2.00	26	900	1700
50	6	6	1.40	0.4	2.20	29	1200	2300
70	12	12	1.40	0.4	2.20	32	1500	3100
95	15	15	1.60	0.4	2.40	37	2000	4200
120	15	18	1.60	0.5	2.40	41	2400	5150
150	15	18	1.80	0.5	2.60	45	2900	6350
185	30	30	2.00	0.6	2.80	50	3600	7900
240	30	34	2.20	0.6	3.00	56	4550	10200
300	30	34	2.40	0.7	3.40	64	5650	12750
400	53	53	2.60	0.7	3.60	70	7000	16100
500	53	53	3.00	0.7	4.00	79	8950	20550
630	53	53	3.40	0.7	4.00	89	11200	26250

* Tabulated Approx. Weight of Cable are only for the purpose of guidline for transportation, loading & unloading purpose.

Cross-sectional view

1. Conductor Material: Aluminium / Copper as per Class-2 of IS 8130
   Shape of Conductor: For Aluminium Conductor - 4 Sq.mm, 6 Sq.mm & 10 Sq. mm Solid/ Stranded Non Compacted Circular,
   16 Sq. mm and above - Stranded Compacted shaped for Copper Conductor - 4 Sq. mm & 6 Sq. mm Solid / Stranded Non Compacted Circular,
   10 Sq. mm - Stranded Compacted Circular, 16 Sq.mm and above - Stranded Compacted Shaped
2. Insulation Material: PVC Type-A or HR PVC Type -C as per IS 5831; Colour : Red, Yellow & Blue
3. Inner Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS Type
4. Outer Sheath: PVC Type ST-1 as per IS 5831; Options: PVC Type ST-2/FR Type FRLS type, LSOH
   Colour of Outer Sheath: Black or any other Colour as per requirement

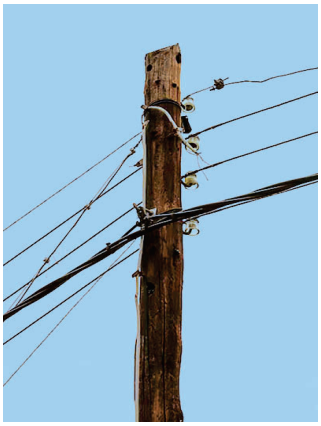
ELECTRICAL PARAMETER

Size (Cross Sectional Area)	Max. Conductor D.C. Resistance at 20 ^{0}C		Approx. Conductor A.C. Resistance at 70 ^{0}C		Reactance of Cable at 50 Hz (Approx)	Capacitance of Cable (Approx)	Normal Current Rating						Short Circuit Current Rating for 1s Duration	
							For Aluminium Conductor			For Aluminium Conductor				
	Aluminium	Copper	Aluminium	Copper			Ground	Duct	Air	Ground	Duct	Air	Aluminium	Copper
Sq. mm	Ω/Km	Ω/Km	Ω/Km	Ω/Km	Ω/Km	µF/Km	A	A		A	A	A	kA	kA
4	7.4100	4.6100	8.8900	5.5300	0.098	0.23	28	23	23	36	30	30	0.304	0.460
6	4.6100	3.0800	5.5300	3.7000	0.096	0.28	35	30	30	45	38	39	0.456	0.690
10	3.0800	1.8300	3.7000	2.2000	0.091	0.34	46	39	40	60	50	52	0.760	1.150
16	1.9100	1.1500	2.2900	1.3800	0.085	0.40	60	50	51	77	64	66	1.220	1.840
25	1.2000	0.7270	1.4400	0.8700	0.083	0.42	76	63	70	99	81	90	1.900	2.880
35	0.8680	0.5240	1.0400	0.6300	0.082	0.48	92	77	86	120	99	110	2.660	4.030
50	0.6410	0.3870	0.7690	0.4640	0.082	0.49	110	95	105	145	125	135	3.800	5.750
70	0.4430	0.2680	0.5320	0.3220	0.076	0.56	135	115	130	175	150	165	5.320	8.050
95	0.3200	0.1930	0.3840	0.2320	0.076	0.58	165	140	155	210	175	200	7.220	10.900
120	0.2530	0.1530	0.3040	0.1840	0.075	0.63	185	155	180	240	195	230	9.120	13.800
150	0.2060	0.1240	0.2470	0.1488	0.074	0.63	210	175	205	270	22	265	11.400	17.300
185	0.1640	0.0991	0.1970	0.1189	0.074	0.64	235	200	240	300	255	305	14.100	21.300
240	0.1250	0.0754	0.1510	0.0912	0.073	0.67	275	235	280	345	295	355	18.200	27.300
300	0.1000	0.0601	0.1220	0.0733	0.073	0.68	305	260	315	385	335	400	22.800	34.500
400	0.0778	0.0470	0.0961	0.0580	0.072	0.70	335	290	375	425	360	455	30.400	46.000
500	0.0605	0.0366	0.0759	0.0459	0.072	0.70	370	320	425	470	390	520	38.000	57.500
630	0.0469	0.0283	0.0610	0.0368	0.072	0.70	405	350	480	555	470	675	47.900	72.500

Note: Normal Current ratings are given in standard conditions, if site conditions are diffferent , current rating should be multipled by rating factor.

AERIAL BUNCHED CABLE (ABC)

AERIAL BUNCHED CABLES (ABC) is a very novel concept for over head power distribution. When compared to the conventional bare conductor over head distribution system, ABC provides higher safety and reliability, low power losses and ultimate system economy by reducing installation, maintenance and operative cost. This system is ideal for rural distribution and attractive for installation in difficult terrains such as hilly areas, forest areas, coastal areas etc. Aerial Bunched Cahles is also considered to be the best choice for power distribution in congested urban areas with narrow lanes and by lanes. in developing urban complex, Aerial Bunched Cables is also considered to be the best choice for power distribution in congested urban areas with narrow lanes and by lanes. In developing urban complex, Aerial Bunched Cables is a better choice because of flexibility for rerouting as demanded by changes in urban development plan.



REF SPECIFICATION : IS-14255, IS 8130 | IEC 60502 Part 1

Phase Conductor (Aluminium)			Street Lighting Conductor (Aluminium)			Messenger Conductor (Aluminium Alloy)		
Nominal Area	Max Conductor D.C. Resistance at 20 ^{0}C	Nominal Thickness of Insulation	Noinal Area	Conductor D.C. Resistance at 20^{0}C	Nominal Thickness of Insulation	Nominal Area	Conductor D.C. Resistance at 20^{0}C	Minimum Breaking Load
Sq. mm	Ω/Km	mm	Sq. mm	Ω/Km	mm	Sq. mm	Ω/Km	kN
16	1.910	1.20	16	1.910	1.20	25	1.380	7.00
25	1.200	1.20	16	1.910	1.20	25	1.380	7.00
35	0.868	1.20	16	1.910	1.20	25	1.380	7.00
50	0.641	1.50	16	1.910	1.20	35	0.986	9.80
70	0.443	1.50	16	1.910	1.20	50	0.689	14.00
95	0.320	1.50	16	1.910	1.20	70	0.492	19.70



XLPE CABLE - SINGLE CORE

Single Core XLPE Insulated Armoured & Unarmoured Cable with Aluminium / Copper Conductor Confirming to IS 7098 (Part -1):1988

Physical Data

Nominal Area	Minium No. of Strand in Conductor		Insulation Thickness		Armour Wire Dimension		Outer Sheath Thickness		Approx. Overall Diameter		Approx. Net Weight of the Cables			
			Arm	Un-Arm	Round	Formed	Arm	Un-Arm	Arm	Un-Arm	Armoured		UnArmoured	
Sq. mm	Aluminiu	Copper	mm	mm	mm	mm	mm	mm	mm	mm	Kg/Km	Kg/Km	Kg/Km	Kg/Km
	Nos	Nos									Al	Cu	Al	Cu
4	1/3	1/7	1.00	0.70	1.40	--	1.24	1.80	10.00	8.00	130	155	75	100
6	1/3	1/7	1.00	0.70	1.40	--	1.24	1.80	11.00	9.00	145	180	90	120
10	1/7	1/7/6	1.00	0.70	1.40	--	1.24	1.80	11.50	9.50	170	230	105	165
16	6	6	1.00	0.70	1.40	--	1.24	1.80	12.50	10.50	200	295	130	225
25	6	6	1.20	0.90	1.40	--	1.24	1.80	14.00	12.00	260	410	180	325
35	6	6	1.20	0.90	1.40	--	1.24	1.80	15.00	13.00	305	515	215	425
50	6	6	1.30	1.00	1.40	--	1.24	1.80	16.50	14.50	365	650	265	550
70	12	12	1.40	1.10	1.40	--	1.24	1.80	18.50	16.50	460	870	350	760
95	15	15	1.40	1.10	--	4.0x0.8	1.40	1.80	19.50	18.00	515	1080	440	1005
120	15	18	1.50	1.20	--	4.0x0.8	1.40	1.80	21.50	19.50	610	1330	530	1245
150	15	18	1.70	1.40	--	4.0x0.8	1.40	2.00	23.00	21.50	715	1605	650	1535
185	30	30	1.90	1.60	--	4.0x0.8	1.40	2.00	25.00	24.00	865	1965	790	1890
240	30	34	2.00	1.70	--	4.0x0.8	1.40	2.00	28.00	26.50	1070	2505	985	2415
300	30	34	2.10	1.80	--	4.0x0.8	1.56	2.00	30.00	28.50	1300	3115	1180	2995
400	53	53	2.40	2.00	--	4.0x0.8	1.56	2.20	34.00	32.50	1610	3945	1495	3835
500	53	53	2.60	2.20	--	4.0x0.8	1.56	2.20	37.50	36.00	1980	4970	1860	4845
630	53	53	2.80	2.40	--	4.0x0.8	1.72	2.20	41.50	40.00	2475	6360	2930	6195
800	53	53	3.10	2.60	--	4.0x0.8	1.72	2.40	47.00	45.50	3095	8055	2930	7890
1000	53	53	3.30	2.80	--	4.0x0.8	1.88	2.60	52.00	50.50	3820	9885	3645	9710

Electrical Data

Nominal Conductor Data	Max. DC Resisance of Conductor at 20^{0}C		Approx. AC Resistance of Max Operating Temp 90^{0}C		Appox. Capacitance		Aprox. Reactance at 50 Hz		Current Rating						Short Circuit Rating for 1 Sec	
									In Ground		In Duct		In Air			
Sq. mm	Ω/Km	Ω/Km	Ω/Km	Ω/Km	Un-Arm	Arm	Un-Arm	Arm	Amps	Amps	Amps	Amps	Amps	Amps	kA (rms)	
	Al	Cu	Al	Cu	µF/Km	µF/Km	Ω/Km	Ω/Km	Al	Cu	Al	Cu	Al	Cu	Al	Cu
4	7.4100	4.6100	9.4848	5.9008	0.29		0.132		37	47	34	43	33	41	0.38	0.57
6	4.6100	3.0800	5.9008	3.9424	0.34		0.123		47	58	43	53	43	52	0.56	0.86
10	3.0800	1.8300	3.9424	2.3424	0.43	0.32	0.114	0.134	59	77	54	70	55	71	0.94	1.43
16	1.9100	1.1500	2.4448	1.472	0.51	0.38	0.108	0.125	76	98	69	89	72	94	1.50	2.29
25	1.2000	0.7270	1.5360	0.93056	0.49	0.38	0.103	0.120	98	126	89	114	98	126	2.35	3.58
35	0.8680	0.5240	1.1110	0.67072	0.57	0.44	0.099	0.140	116	150	106	136	119	154	3.29	5.01
50	0.6410	0.3870	0.8205	0.49536	0.58	0.46	0.094	0.108	137	177	124	160	145	187	4.70	7.15
70	0.4400	0.2680	0.5632	0.34304	0.63	0.51	0.090	0.102	168	216	151	195	185	238	6.58	10.01
95	0.3200	0.1930	0.4096	0.24704	0.73	0.59	0.087	0.100	202	260	181	233	235	303	8.93	13.59
120	0.2530	0.1530	0.3238	0.19584	0.74	0.61	0.084	0.097	230	295	206	264	276	354	11.28	17.16
150	0.2060	0.1240	0.2637	0.15872	0.73	0.61	0.084	0.094	256	329	229	294	314	403	14.10	21.45
185	0.1640	0.0991	0.2099	0.126848	0.69	0.59	0.084	0.093	290	371	258	330	366	468	17.39	26.46
240	0.1250	0.0754	0.1600	0.096512	0.74	0.64	0.081	0.090	335	427	298	379	434	553	22.56	34.32
300	0.1000	0.0601	0.1280	0.076928	0.80	0.69	0.080	0.088	376	477	333	422	500	634	28.20	42.90
400	0.0778	0.0470	0.0996	0.06016	0.83	0.70	0.079	0.087	429	537	378	473	589	737	37.60	57.20
500	0.0605	0.0366	0.0774	0.046848	0.83	0.71	0.078	0.086	485	598	426	525	658	844	47.00	71.50
630	0.0469	0.0283	0.0600	0.036224	0.87	0.75	0.079	0.084	546	661	477	578	793	961	59.22	90.09
800	0.0367	0.0221	0.0470	0.028288	0.92	0.78	0.076	0.083	608	721	528	626	907	1077	75.20	114.40



XLPE CABLE - TWO CORE

Two Core XLPE Insulated Armoured & Unarmoured Cable with Aluminium / Copper Conductor Confirming to IS 7098 (Part -1):1988

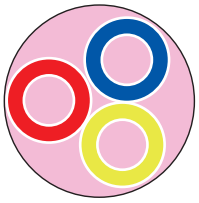
Physical Data

Nominal Area	Minium No. of Strand in Conductor		Insulation Thickness (Nominal)	Minimum inner sheath thickness	Armour Wire Dimension		Outer Sheath Thickness		Approx. Overall Diameter		Approx. Net Weight of the Cables			
					Round	Formed	Arm	Un-Arm	Arm	Un-Arm	Armoured		UnArmoured	
Sq. mm	Aluminium	Copper	mm	mm	mm	mm	mm	mm	mm	mm	Kg/Km	Kg/Km	Kg/Km	Kg/Km
	Nos	Nos									Al	Cu	Al	Cu
4	1/3	1/7	0.70	0.30	1.40	--	1.24	1.80	14.50	13.00	405	450	145	195
6	1/3	1/7	0.70	0.30	1.40	--	1.24	1.80	16.00	14.00	455	525	175	245
10	1/7	1/7/6	0.70	0.30	1.40	--	1.24	1.80	17.50	16.00	540	660	215	335
16	6	6	0.70	0.30	1.40	--	1.40	1.80	16.50	15.00	555	750	250	440
25	6	6	0.90	0.30	--	4.0X0.8	1.40	2.00	18.50	18.00	575	875	355	655
35	6	6	0.90	0.30	--	4.0x0.8	1.40	2.00	20.00	19.50	685	1110	440	865
50	6	6	1.00	0.30	--	4.0x0.8	1.40	2.00	22.00	21.50	810	1380	545	1115
70	12	12	1.10	0.30	--	4.0x0.8	1.56	2.00	25.50	24.50	1040	1860	710	1535
95	15	15	1.10	0.40	--	4.0x0.8	1.56	2.20	28.00	27.50	1275	2425	930	2070
120	15	18	1.20	0.40	--	4.0x0.8	1.56	2.20	30.50	30.00	1500	3945	1110	2555
150	15	18	1.40	0.40	--	4.0x0.8	1.72	2.20	33.00	32.00	1750	3540	1320	3105
185	30	30	1.60	0.50	--	4.0x0.8	1.72	2.40	36.00	35.50	2085	4310	16445	3970
240	30	34	1.70	0.50	--	4.0x0.8	1.88	2.60	40.50	40.00	2585	5475	2085	4975
300	30	34	1.80	0.60	--	4.0x0.8	2.04	2.80	43.50	43.50	3100	6770	2555	6220
400	53	53	2.00	0.60	--	4.0x0.8	2.36	3.00	49.50	59.00	3860	8580	3195	9715
500	53	53	2.20	0.70	--	4.0x0.8	2.52	3.40	54.00	54.00	4725	10755	4055	10085
630	53	53	2.40	0.70	--	4.0x0.8	2.68	3.60	60.00	60.00	5775	13635	5035	12895

Electrical Data

Nominal Conductor Data	Max. DC Resisance of Conductor at 20°C		Approx. AC Resistance of Max Operating Temp 90°C		Appox. Capacitance	Aprox. Reactance at 50 Hz	Current Rating						Short Circuit Rating for 1 Sec		Volrage Drop	
							In Ground		In Duct		In Air					
Sq. mm	Ω/Km	Ω/Km	Ω/Km	Ω/Km			Amps	Amps	Amps	Amps	Amps	Amps	kA (rms)		V/A/Km	
	Al	Cu	Al	Cu	µF/Km	Ω/Km	Al	Cu	Al	Cu	Al	Cu	Al	Cu	Al	Cu
4	7.4100	4.6100	9.4848	5.9008	0.220	0.093	42	54	36	45	38	48	0.38	0.57	16.459	10.186
6	4.6100	3.0800	5.9008	3.9424	0.250	0.890	55	67	46	56	50	61	0.56	0.86	10.239	6.809
10	3.0800	1.8300	3.9424	2.3424	0.310	0.084	68	89	57	75	64	83	0.94	1.43	6.842	4.038
16	1.9100	1.1500	2.4448	1.472	0.360	0.081	89	115	74	96	83	108	1.50	2.29	4.194	2.550
25	1.2000	0.7270	1.5360	0.93056	0.410	0.081	114	147	95	122	109	140	2.35	3.58	2.671	1.612
35	0.8680	0.5240	1.1110	0.67072	0.470	0.079	136	176	113	146	133	172	3.29	5.01	1.927	1.165
50	0.6410	0.3870	0.8205	0.49536	0.500	0.078	161	208	134	173	162	208	4.70	7.15	1.430	0.866
70	0.4400	0.2680	0.5632	0.34304	0.530	0.074	197	253	164	211	204	262	6.58	10.01	0.992	0.606
95	0.3200	0.1930	0.4096	0.24704	0.610	0.073	235	302	196	252	251	322	8.93	13.59	0.723	0.446
120	0.2530	0.1530	0.3238	0.19584	0.630	0.072	266	340	222	284	287	368	11.28	17.16	0.577	0.363
150	0.2060	0.1240	0.2637	0.15872	0.640	0.073	296	379	248	317	328	419	14.10	21.45	0.476	0.304
185	0.1640	0.0991	0.2099	0.126848	0.650	0.072	335	425	281	357	379	482	17.39	26.46	0.386	0.254
240	0.1250	0.0754	0.1600	0.096512	0.660	0.071	385	486	324	409	448	566	22.56	34.32	0.307	0.211
300	0.1000	0.0601	0.1280	0.076928	0.670	0.071	432	541	364	456	513	644	28.20	42.90	0.256	0.185
400	0.0778	0.0470	0.0996	0.06016	0.670	0.070	487	602	412	508	593	734	37.60	57.20	0.214	0.165
500	0.0605	0.0366	0.0774	0.046848	0.690	0.070	548	665	463	562	683	831	47.00	71.50	0.185	0.151
630	0.0469	0.0283	0.0600	0.036224	0.730	0.070	612	728	518	616	784	936	59.22	90.09	0.164	0.141

XLPE CABLE - THREE CORE



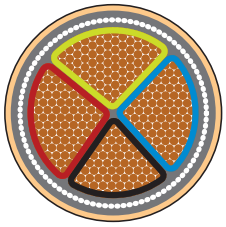
Three Core XLPE Insulated Armoured & Unarmoured Cable with Aluminium / Copper Conductor Confirming to IS 7098 (Part -1):1988

Physical Data

Nominal Area	Minium No. of Strand in Conductor		Insulation Thickness (Nominal)	Minimum inner sheath thickness	Armour Wire Dimension		Outer Sheath Thickness		Approx. Overall Diameter		Approx. Net Weight of the Cables			
					Round	Formed	Arm	Un-Arm	Arm	Un-Arm	Armoured		UnArmoured	
Sq. mm	Aluminium	Copper	mm	mm	mm	mm	mm	mm	mm	mm	Kg/Km	Kg/Km	Kg/Km	Kg/Km
	Nos	Nos									Al	Cu	Al	Cu
4	1/3	1/7	0.70	0.30	1.40	--	1.24	1.80	15.00	13.50	445	520	170	245
6	1/3	1/7	0.70	0.30	1.40	--	1.24	1.80	17.00	14.50	505	615	210	315
10	1/7	1/7/6	0.70	0.30	1.40	--	1.24	1.80	19.00	16.50	605	790	260	440
16	6	6	0.70	0.30	--	4.0X0.8	1.24	1.80	18.00	17.00	560	850	335	625
25	6	6	0.90	0.30	--	4.0X0.8	1.40	2.00	21.00	20.00	720	1170	480	925
35	6	6	0.90	0.30	--	4.0x0.8	1.40	2.00	23.00	22.00	890	1525	600	1235
50	6	6	1.00	0.30	--	4.0x0.8	1.40	2.00	25.00	24.50	1055	1910	750	1605
70	12	12	1.10	0.40	--	4.0x0.8	1.56	2.20	30.00	28.50	1400	2635	1030	2265
95	15	15	1.10	0.40	--	4.0x0.8	1.56	2.20	32.00	31.00	1705	3415	1290	3000
120	15	18	1.20	0.40	--	4.0x0.8	1.56	2.20	35.00	34.00	2015	4185	1560	3725
150	15	18	1.40	0.50	--	4.0x0.8	1.72	2.20	39.00	38.00	2415	5095	1920	4600
185	30	30	1.60	0.50	--	4.0x0.8	1.88	2.20	43.00	42.00	2930	6265	2375	5710
240	30	34	1.70	0.60	--	4.0x0.8	2.04	2.80	49.00	47.50	3635	7975	3025	7365
300	30	34	1.80	0.60	--	4.0x0.8	2.04	2.80	49.00	47.50	3635	7975	3025	7365
400	53	53	2.00	0.70	--	4.0x0.8	2.50	3.20	60.00	58.50	5450	1250	4660	11740
500	53	53	2.20	0.70	--	4.0x0.8	2.68	3.60	67.00	65.50	6720	15760	5885	14930
630	53	53	2.40	0.70	--	4.0x0.8	2.84	3.80	74.00	72.50	8220	20005	7320	19105

Electrical Data

Nominal Conductor Data	Max. DC Resisance of Conductor at 20°C		Approx. AC Resistance of Max Operating Temp 90°C		Appox. Capacitance	Aprox. Reactance at 50 Hz	Current Rating						Short Circuit Rating for 1 Sec		Volrage Drop	
							In Ground		In Duct		In Air					
Sq. mm	Ω/Km	Ω/Km	Ω/Km	Ω/Km			Amps	Amps	Amps	Amps	Amps	Amps	kA (rms)		V/A/Km	
	Al	Cu	Al	Cu	µF/Km	Ω/Km	Al	Cu	Al	Cu	Al	Cu	Al	Cu	Al	Cu
4	7.4100	4.6100	9.4848	5.9008	0.065	0.093	35	45	30	38	32	41	0.38	0.57	16.459	10.186
6	4.6100	3.0800	5.9008	3.9424	0.071	0.089	46	56	38	47	42	52	0.56	0.86	10.239	6.809
10	3.0800	1.8300	3.9424	2.3424	0.081	0.084	57	74	48	62	54	70	0.94	1.43	6.842	4.038
16	1.9100	1.1500	2.4448	1.472	0.088	0.081	74	95	61	79	69	89	1.50	2.29	4.194	2.550
25	1.2000	0.7270	1.5360	0.93056	0.089	0.081	95	122	79	102	93	119	2.35	3.58	2.671	1.612
35	0.8680	0.5240	1.1110	0.67072	0.096	0.079	114	146	94	122	114	147	3.29	5.01	1.927	1.165
50	0.6410	0.3870	0.8205	0.49536	0.098	0.078	134	173	112	144	138	179	4.70	7.15	1.430	0.866
70	0.4400	0.2680	0.5632	0.34304	0.100	0.074	164	212	137	177	175	226	6.58	10.01	0.992	0.606
95	0.3200	0.1930	0.4096	0.24704	0.110	0.073	197	254	164	212	216	279	8.93	13.59	0.723	0.446
120	0.2530	0.1530	0.3238	0.19584	0.110	0.072	223	287	187	240	249	320	11.28	17.16	0.577	0.363
150	0.2060	0.1240	0.2637	0.15872	0.110	0.073	249	321	209	269	284	365	14.10	21.45	0.476	0.304
185	0.1640	0.0991	0.2099	0.126848	0.110	0.072	282	362	238	304	329	422	17.39	26.46	0.386	0.254
240	0.1250	0.0754	0.1600	0.096512	0.110	0.071	327	418	276	352	392	500	22.56	34.32	0.307	0.211
300	0.1000	0.0601	0.1280	0.076928	0.120	0.071	369	469	312	696	452	574	28.20	42.90	0.256	0.185
400	0.0778	0.0470	0.0996	0.06016	0.120	0.070	420	528	356	447	526	664	37.60	57.20	0.214	0.165
500	0.0605	0.0366	0.0774	0.046848	0.120	0.070	478	593	412	511	612	760	47.00	71.50	0.185	0.151
630	0.0469	0.0283	0.0600	0.036224	0.120	0.070	542	661	468	571	712	870	59.22	90.09	0.164	0.141



XLPE CABLE - THREE & HALF CORE

Three & Half Core XLPE Insulated Armoured & Unarmoured Cable with Aluminium / Copper Conductor Confirming to IS 7098 (Part -1):1988

Physical Data

Nominal Area		Minium No. of Strand in Conductor		Insulation Thickness (Nominal)		Inner Sheath Thickness (Min)	Armour Dimension	Outer Sheath Thickness		Approx. Overall Diameter		Approx. Net Weight of the Cables			
				Power	Neutral		Formed	Arm	Un-Arm	Arm	Un-Arm	Armoured		UnArmoured	
Sq. mm	Sq. mm	Aluminium	Copper	Sq. mm	Sq. mm	mm	mm	mm	mm	mm	mm	Kg/Km	Kg/Km	Kg/Km	Kg/Km
		Nos	Nos									Al	Cu	Al	Cu
25	16	6	6	0.90	0.70	0.30	4.0x0.8	1.40	2.00	22.50	21.50	805	1350	540	1080
35	16	6	6	0.90	0.70	0.30	4.0x0.8	1.40	2.00	24.50	35.50	980	1705	650	1380
50	25	6	6	1.00	0.90	0.30	4.0x0.8	1.40	2.00	27.00	26.00	1200	2205	830	1835
70	35	12	12	1.10	0.90	0.400	4.0x0.8	1.56	2.20	31.00	30.00	1550	2995	1145	1590
95	50	15	15	1.10	1.00	0.400	4.0X0.8	1.56	2.20	31.50	33.50	925	3920	1455	3450
120	70	15	18	1.20	1.10	0.400	4.0x0.8	1.72	2.20	38.50	37.00	2335	4910	1795	4370
150	70	15	18	1.40	1.10	0.500	4.0x0.8	1.72	2.40	42.00	41.00	2730	5820	2165	5855
185	95	30	30	1.60	1.10	0.500	4.0x0.8	1.88	2.60	46.50	45.50	3325	7230	2700	6610
240	120	30	34	1.70	1.20	0.600	4.0x0.8	2.04	2.80	52.50	51.50	4115	9180	3445	8505
300	150	30	34	1.80	1.40	0.600	4.0x0.8	2.20	3.00	87.50	86.50	4960	11350	4205	10595
400	185	53	53	2.00	1.60	0.700	4.0x0.8	2.52	3.40	65.50	65.00	6230	14420	5365	13560
500	240	53	53	2.20	1.70	0.700	4.0x0.8	2.68	3.60	72.50	72.00	7655	18145	6725	17215
630	300	53	53	2.40	1.80	0.700	4.0x0.8	3.00	4.00	82.00	81.00	9485	23105	8465	22085

Electrical Data

Nominal Conductor Data	Max. DC Resisance of Conductor at 20°C		Approx. AC Resistance of Max Operating Temp 90°C		Appox. Capacitance	Aprox. Reactance at 50 Hz	Current Rating						Short Circuit Rating for 1 Sec		Volrage Drop	
							In Ground		In Duct		In Air					
Sq. mm	Ω/Km	Ω/Km	Ω/Km	Ω/Km			Amps	Amps	Amps	Amps	Amps	Amps	kA (rms)		V/A/Km	
	Al	Cu	Al	Cu	µF/Km	Ω/Km	Al	Cu	Al	Cu	Al	Cu	Al	Cu	Al	Cu
25	1.2000	0.7270	1.5360	0.93056	0.089	0.081	95	122	79	102	93	119	2.35	3.58	2.671	
35	0.8680	0.5240	1.1110	0.67072	0.096	0.079	114	146	94	122	114	147	3.29	5.01	1.927	1.165
50	0.6410	0.3870	0.8205	0.49536	0.098	0.078	134	173	112	144	138	179	4.70	7.15	1.430	0.866
70	0.4400	0.2680	0.5632	0.34304	0.100	0.074	164	212	137	177	175	226	6.58	10.01	0.992	0.606
95	0.3200	0.1930	0.4096	0.24704	0.110	0.073	197	254	164	212	216	279	8.93	13.59	0.723	0.446
120	0.2530	0.1530	0.3238	0.19584	0.110	0.072	223	287	187	240	249	320	11.28	17.16	0.577	0.363
150	0.2060	0.1240	0.2637	0.15872	0.110	0.073	249	321	209	269	284	365	14.10	21.45	0.476	0.304
185	0.1640	0.0991	0.2099	0.126848	0.110	0.072	282	362	238	304	329	422	17.39	26.46	0.386	0.254
240	0.1250	0.0754	0.1600	0.096512	0.110	0.071	327	418	276	352	392	500	22.56	34.32	0.307	0.211
300	0.1000	0.0601	0.1280	0.076928	0.120	0.071	369	469	312	696	452	574	28.20	42.90	0.256	0.185
400	0.0778	0.0470	0.0996	0.06016	0.120	0.070	420	528	356	447	526	664	37.60	57.20	0.214	0.165
500	0.0605	0.0366	0.0774	0.046848	0.120	0.070	478	593	412	511	612	760	47.00	71.50	0.185	0.151
630	0.0469	0.0283	0.0600	0.036224	0.120	0.070	542	661	468	571	712	870	59.22	90.09	0.164	0.141

Few of our valuable clients

STATE ELECTRICITY BOARD

CENTRAL GOVT. / PSU

OVERSEAS COMPANIES

RENEWABLE ENERGY

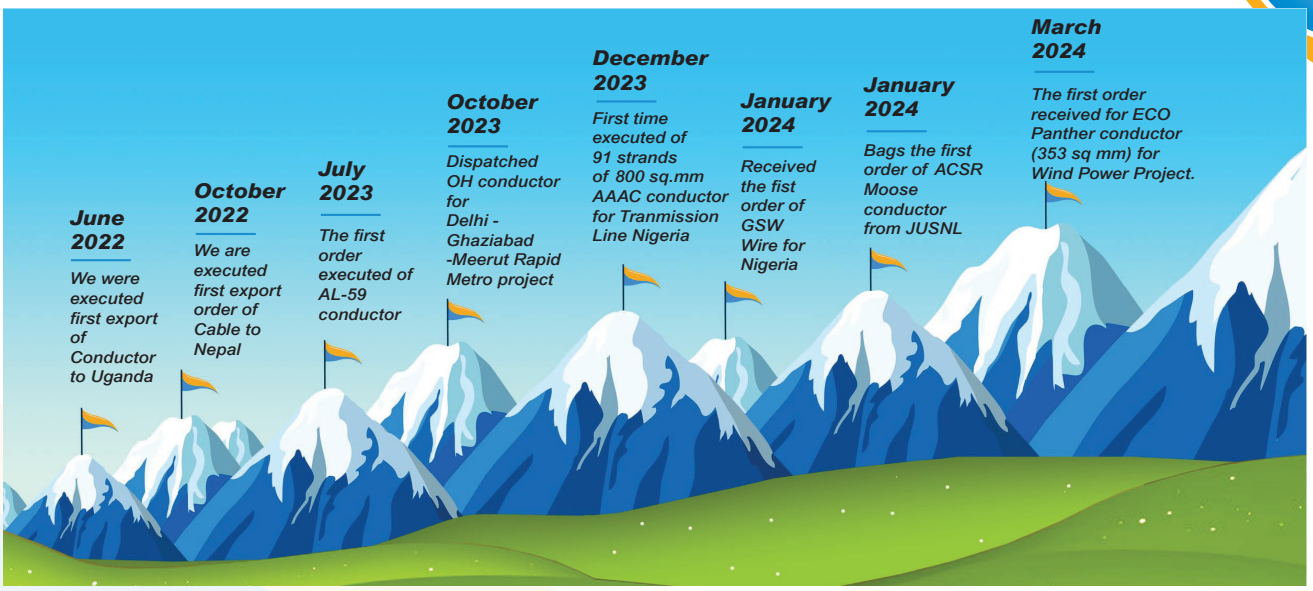
ISO Certificates







Our Achievements



BIS Certificates

SL NO.	LICENCE NO.	PRODUCT DESCRIPTION	IS NO.
1	CM/L-5100057985	PVC INSULATED UNSHEATHED AND SHEATHED UNARMOURED CABLES CORDS WITH RIGID AND FLEXIBLE CONDUCTOR UPTO 1.1KV	694: 2010
2	CM/L-5100042871	PVC INSULATED & COVERD HEAVY DUTY ELECTRIC CABLE UPTO 1.1KV	1554: P-1: 1988
3	CM/L-5100025164	XLPE INSULATED & PVC SHEATHED ARMOURED & UNARMOURED CABLE UPTO 1.1KV	7098: P-1: 1988
4	CM/L-5100025265	LT AERIAL BUNCHED CABLE UPTO 1.1KV	14255 : 1995
5	CM/L-5500129006	ALL ALUMINIUM CONDUCTORS FOR OVER HEAD TRANSMISSION (AAC)	398 P-1: 1996
6	CM/L-5100163681	ALUMINIUM CONDUCTORS WITH STEEL REINFORCED FOR OVER HEAD TRANSMISSION (ACSR)	398 P-2: 1996
7	CM/L-5500129208	ALL ALUMINIUM ALLOY CONDUCTORS FOR OVER HEAD TRANSMISSION (AAAC)	398 P-4: 1994
8	CM/L-5500145004	ALUMINIUM CONDUCTORS WITH GALVANISED STEEL RAIN FORCED FOR OVER HEAD TRANSMISSION FOR 400KV AND ABOVE (ACSR)	399 P-5: 1992
9	CM/L-5500158308	HIGH CONDUCTIVITY ALUMINIUM ALLOY CONDUCTORS FOR OVERHEAD POWER TRANSMISSION PURPOSED, OF SIZES UPTO AND INCLUDING 774 MM^2	398 P-6:2021
10	CM/L-5500138512	HOT DIP GALVANISED STAY STRAND	2141 :2000
11	CM/L- 5500158510	GALVANIZED STRAND FOR EARTHING , CONSTRUCTION - UPTO AND INCLUDING 7 x 4.00 MM, WIRE TENSILE GRADE - 981, 1100 AND 1310	12776:2002

Testing Facilities

- ✓ Breaking load test of conductor
- ✓ Elongation Test of conductor
- ✓ Conductor Resistance Test
- ✓ Tensile Test for Insulation
- ✓ Elongation Test for Insulation
- ✓ Dip Test for Steel wires
- ✓ Ductility Test for Steel wires
- ✓ Insulation Resistance Test
- ✓ Conditioning Chamber Test
- ✓ Hot set Test for XLPE cables
- ✓ Hot Air Oven Schematic Test XLPE cables
- ✓ Water Absorption Test
- ✓ Loss of Mass Test for Cables
- ✓ Thermal stability test of cables
- ✓ Mass of Zinc coating test
- ✓ Resistance Test for cable
- ✓ Smoke Density Test for FRLSH / FR cable
- ✓ Oxygen index for FRLSH / FR cable
- ✓ Halogen Acid Gas Test for FRLSH cable
- ✓ Shrinkage Test for Cables
- ✓ Flame Retardant Test for FR/FRLSH cable
- ✓ Flammability Test for cables
- ✓ Carbon content Test for XLPE
- ✓ Cold Bend Test For Cables
- ✓ Colour fastness test for cables
- ✓ High Voltage Test for cables and more

Regd. Office & Works #

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